



Time Series Knights

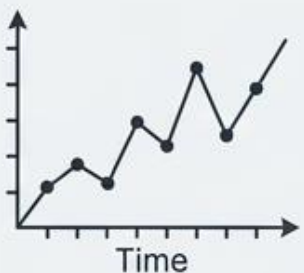


Time Series Analysis

From Demand Forecasting to Causality and Beyond

Chainarong Amornbunchornvej, Ph.D.

NECTEC, Thailand | chai@ieee.org



The Four Questions of Signal Analysis

A time series is a sequence of data points over evenly spaced intervals where shuffling the rows destroys the information.

To understand it, we must apply four distinct lenses.



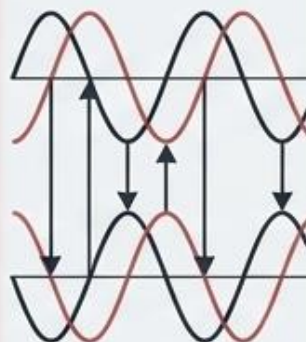
Predict
(Forecasting)

What will
happen?



Cause
(Causality)

Why did it
happen?



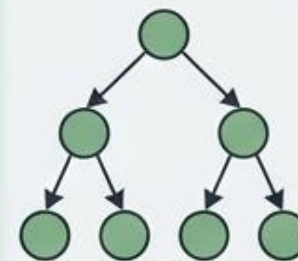
Decompose
(Decomposition)

How is the
signal built?

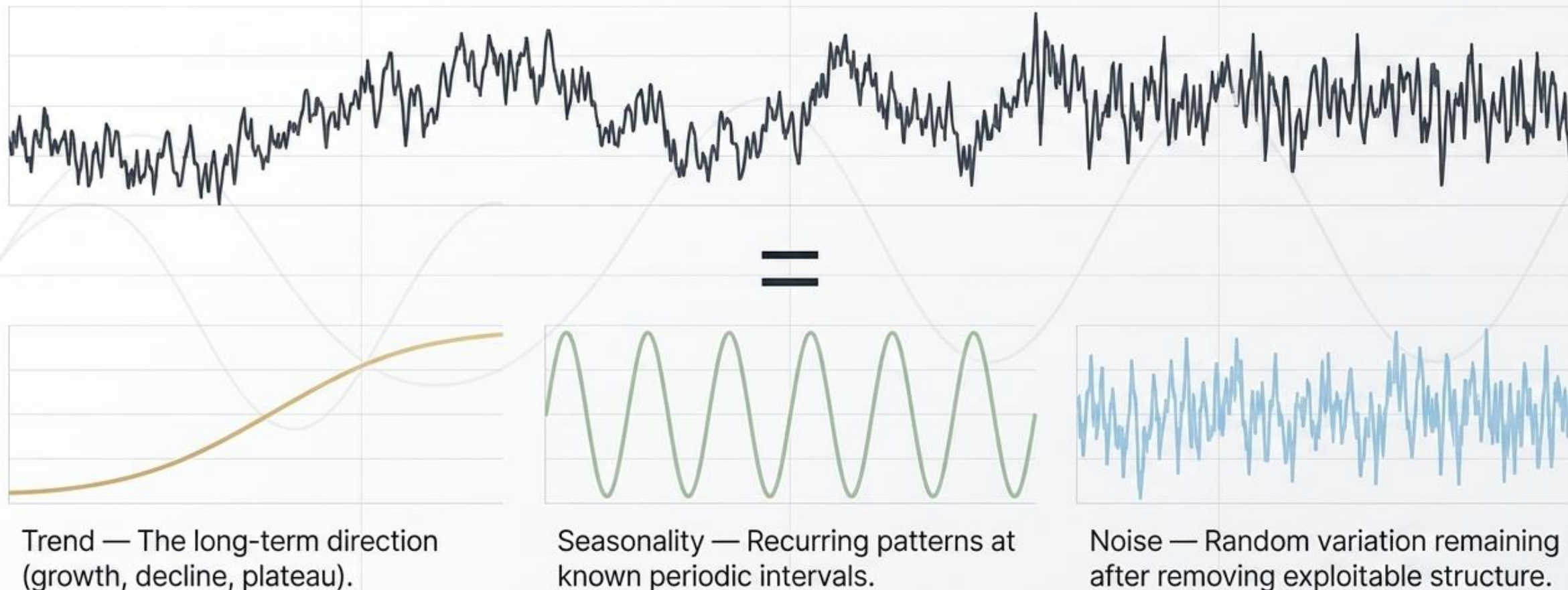


Coordinate
(Coordination)

Who is
leading
whom?



The Anatomy of a Signal



Core assumption: Temporal structure is exploitable information. The past informs the future.

Section 2

Demand Forecasting

From Classical Methods to Deep Learning for Factory Power-Load Prediction

Excess Inventory & Wasted Capacity

Capital becomes tied up in overstock, and energy generation exceeds consumption, leading to costly waste.

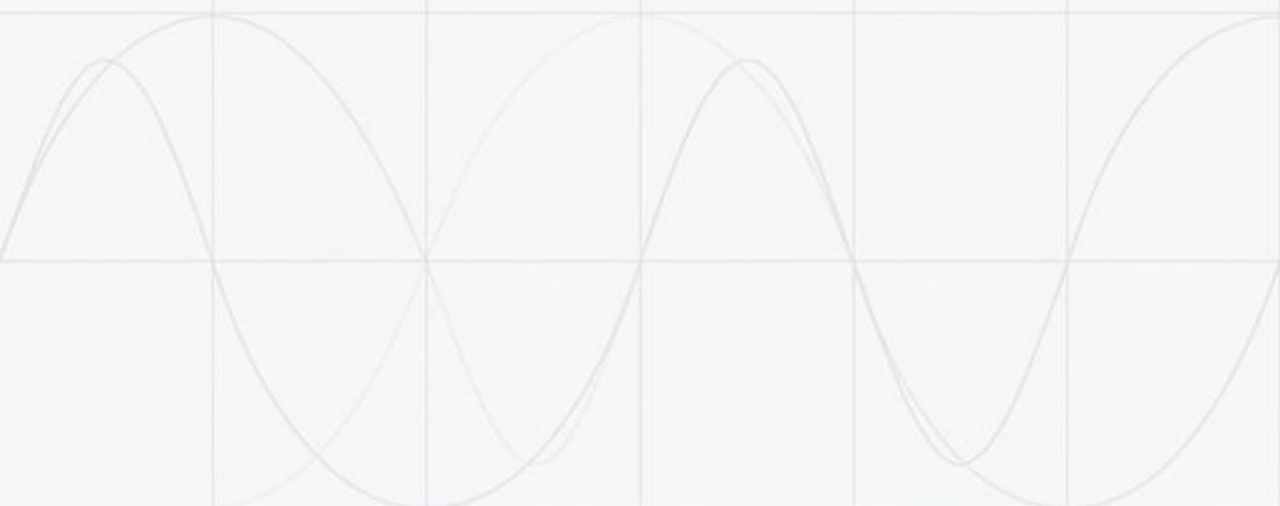


Stockouts & Customer Churn

Lost sales, power blackouts, and unmet demand propagate cascading failures throughout the supply chain.

Forecast quality is a direct proxy for operational efficiency. The penalty for under-predicting rarely equals the penalty for over-predicting.

The Forecasting Optimization Engine



↑

Univariate to Multivariate

Y-Axis

X-Axis

Simple to Complex

LSTMs & Transformers

Deep learning architectures. Use when managing massive datasets with multivariate inputs, exogenous features, and complex, non-linear dependencies.

ARIMA & ETS

Explicitly models lags. Use when data is univariate, stationary-able, and highly interpretable results are strictly required. Ideal for short-horizon retail.

Facebook Prophet

Additive regression at scale. Use when the series is driven by human calendars with strong multiple seasonalities (holidays, weekends) and changepoints.

The Single-Scale Limitation

The Goal: Predict total next-hour electricity demand across unique factories to optimize generator scheduling.

Factory A: The Daily Rhythm



Factory B: The Weekly Wave

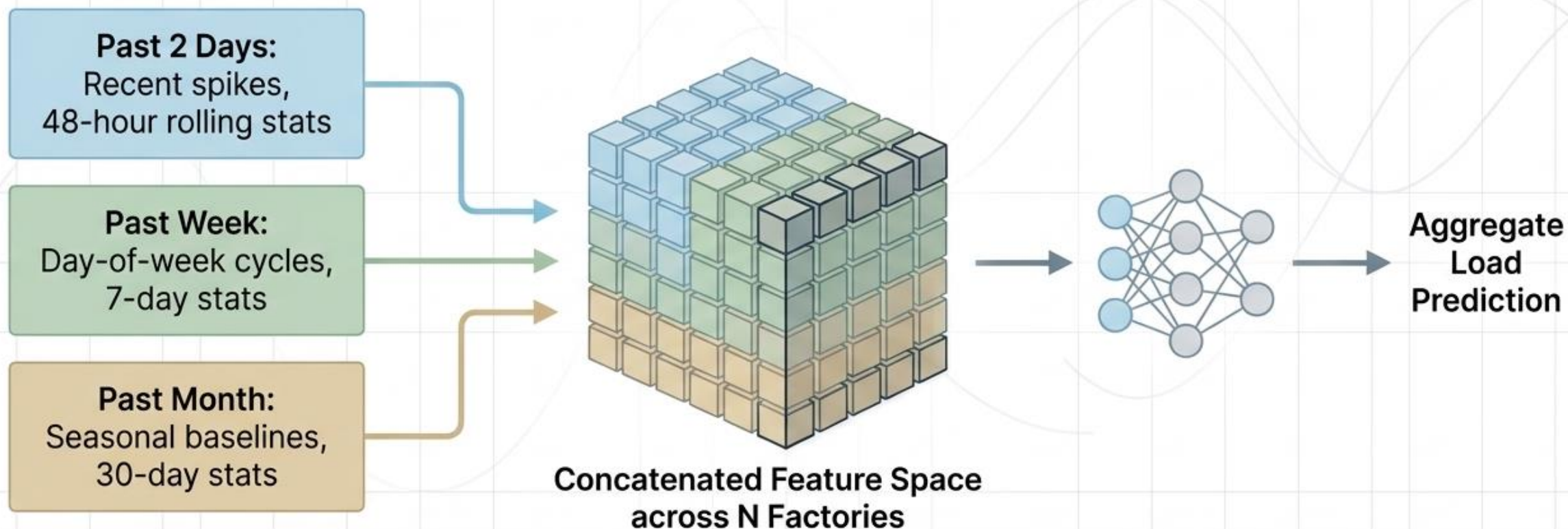


Factory C: The Monthly Quota



A standard forecasting model forced to look at a single historical window will capture one rhythm but obliterate the others

Multi-Temporal Resolution Architecture

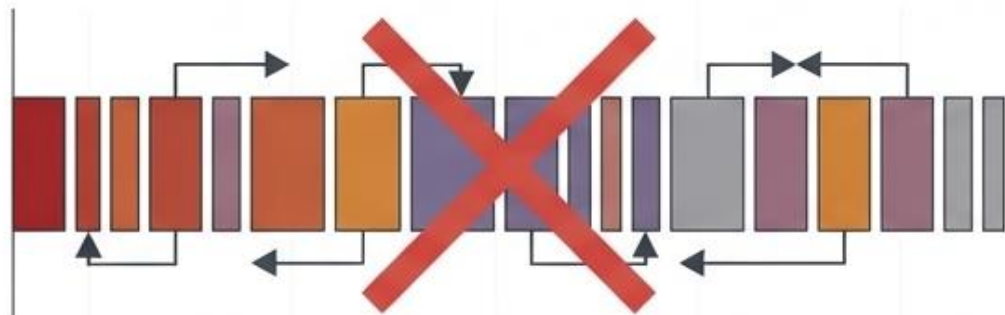


Key Insight

Instead of forcing a single lookback window, the model dynamically learns which temporal scales are structurally important for each specific site.

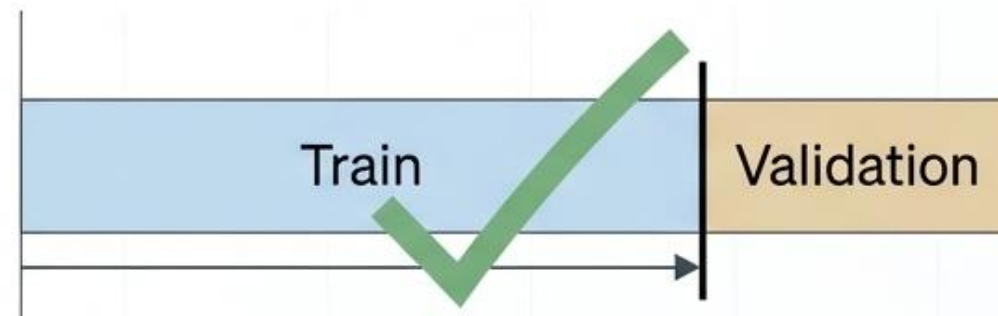
Golden Rule 1: The Validation Trap

Information Leakage



Time series are NOT i.i.d. Randomly splitting data leaks future information into the past, guaranteeing an overfit model that will collapse in production.

Expanding Window Validation



Always split chronologically.
Rolling-origin (walk-forward) validation ensures the model is only ever tested on unseen, future data.

Golden Rule 2: Metric Selection is Policy Definition

Metric	Formula Intuition	When to Use It
Mean Absolute Error (MAE)	Average absolute error; symmetric penalty.	The default baseline. Easy to explain to business stakeholders.
Root Mean Square Error (RMSE)	Squares the errors before averaging, heavily penalizing large misses.	Use when big misses cost disproportionately more than small misses (e.g., severe power blackouts).
Mean Absolute Percentage Error (MAPE)	Error as a percentage of actuals.	Cross-factory or cross-product comparison where scales vary drastically.

The Practitioner's Reality Check



Data Over Algorithms

Algorithms do not fix bad data. No mathematical framework can save a model if the training data is randomized, leaking future states into the past.



Reality Over Mechanics

Beware confounding factors.
Observation is not intervention.
Combining timescales protects against forcing dynamic operations into rigid, single-scale assumptions.



Match the Tool to the System

Forecasting is an optimization lever.
Match the complexity of the method—from simple ETS to Multi-Temporal Neural Networks—to the richness of your data.

Use Case: Factory Power Load Prediction

The Goal

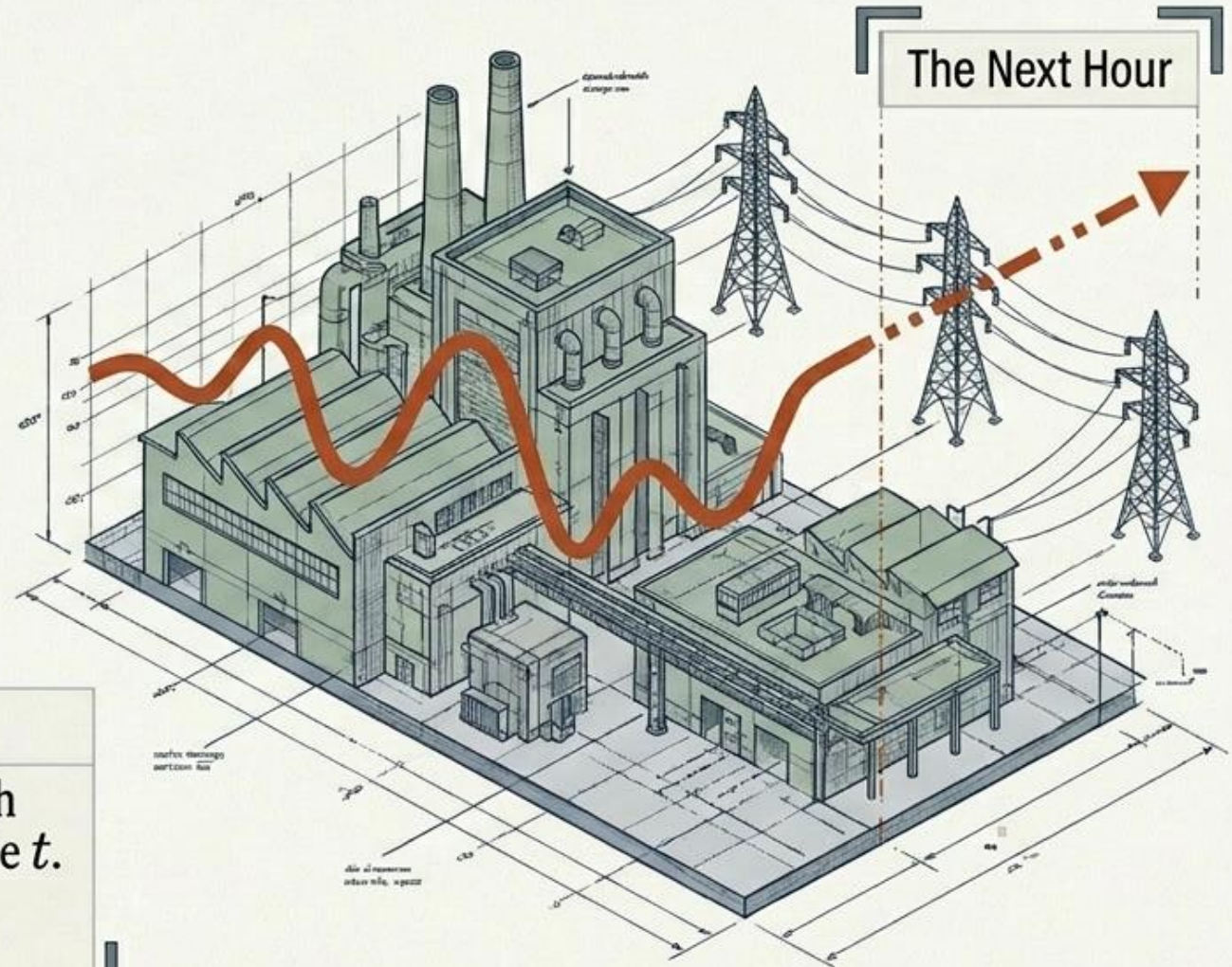
Predict the exact next-hour aggregate electricity demand for continuous factory operations.

The Input Features

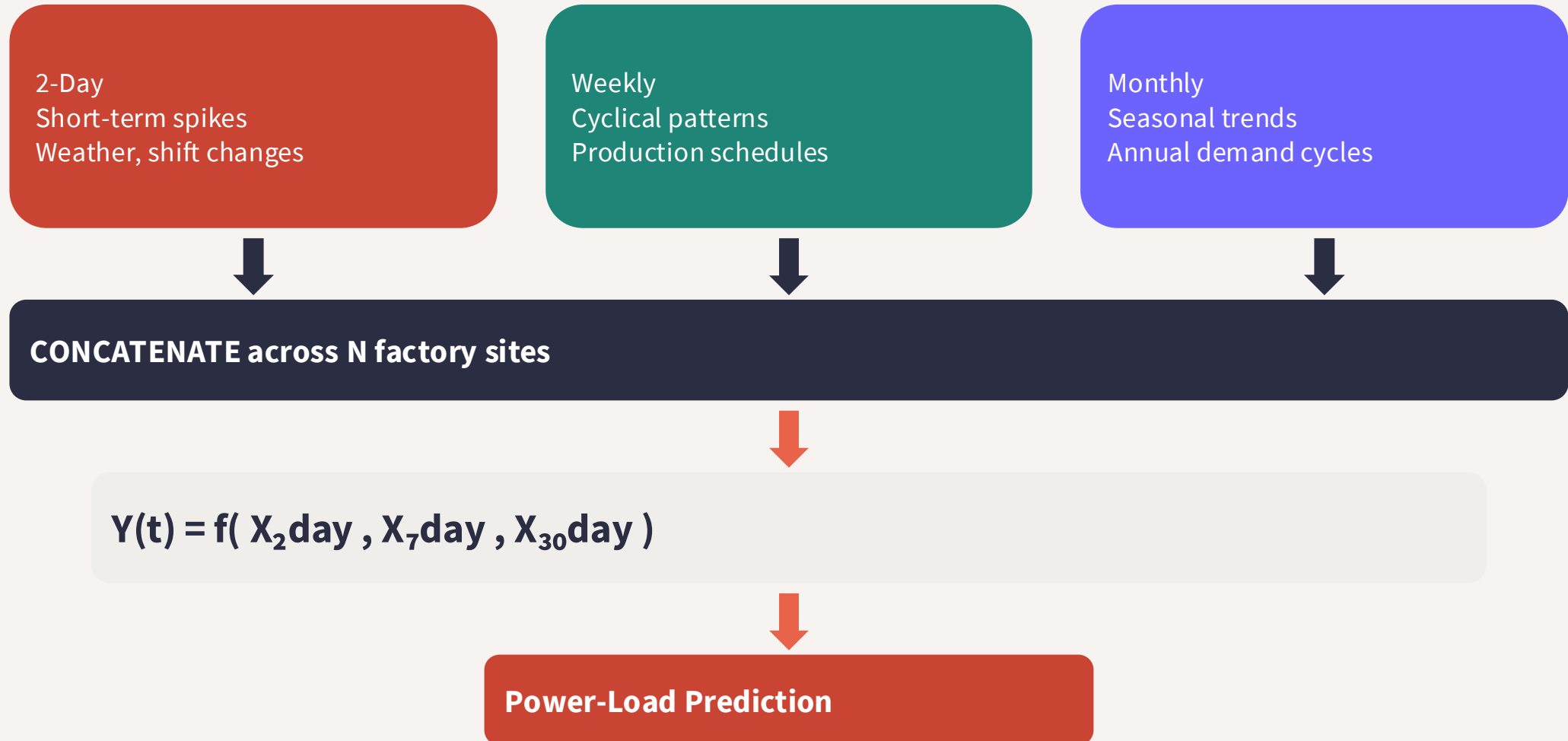
Historical power load, localized temperature data, time of day, and day of the week.

The Business Constraint

The power generator must produce exactly enough electricity to match the total aggregate load at time t . Overproduction wastes capital; underproduction triggers catastrophic equipment failure.



Multi-Temporal Resolution Regression



The Core Challenge: Temporal Complexity

A naive aggregate model fails because it flattens temporal diversity.

The Aggregation Problem:

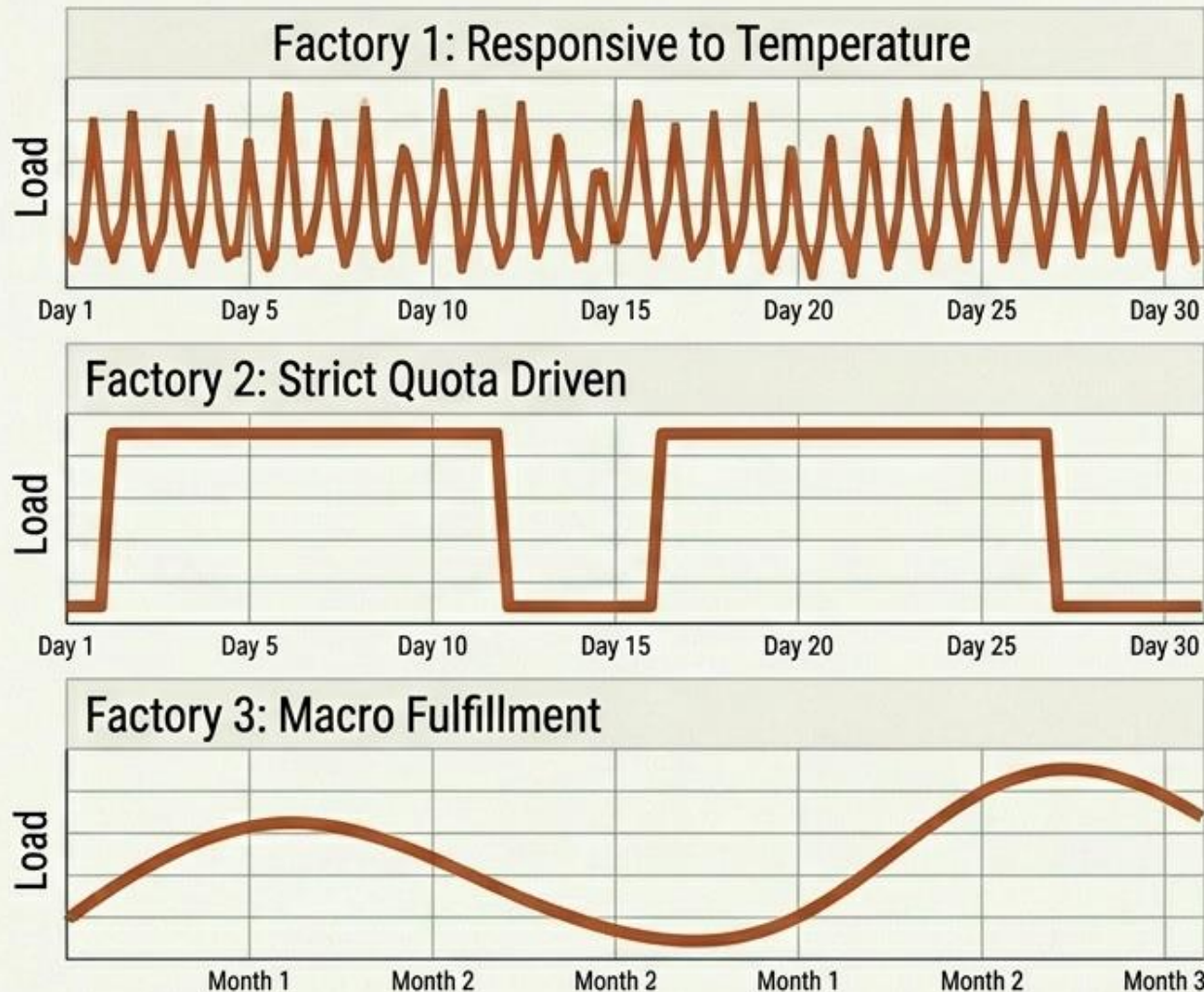
We must predict total load across N different factories simultaneously.

Distinct Heartbeats:

Every factory operates on completely different temporal scales. Factory 1 reacts to daily weather, Factory 2 to weekly schedules, and Factory 3 to monthly supply chains.

The Failure of the Average:

Attempting to predict total power load by flattening these distinct signals into a single, generic historical window destroys the localized signal.



Feature Construction at Three Timescales

X_2^d

2-Day Window

Recent load spikes
Shift patterns
Hour-of-day effects

X_{wk}

Weekly Window

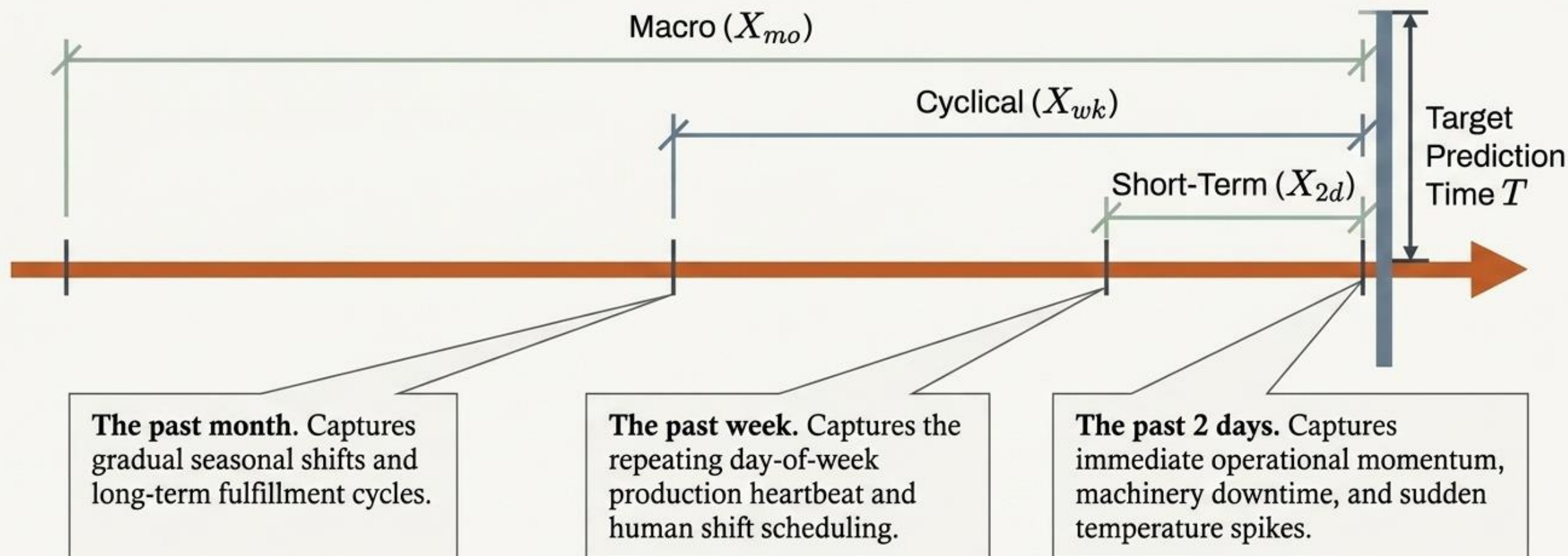
Day-of-week cycles
Weekend vs weekday
Short-term weather

X_{mo}

Monthly Window

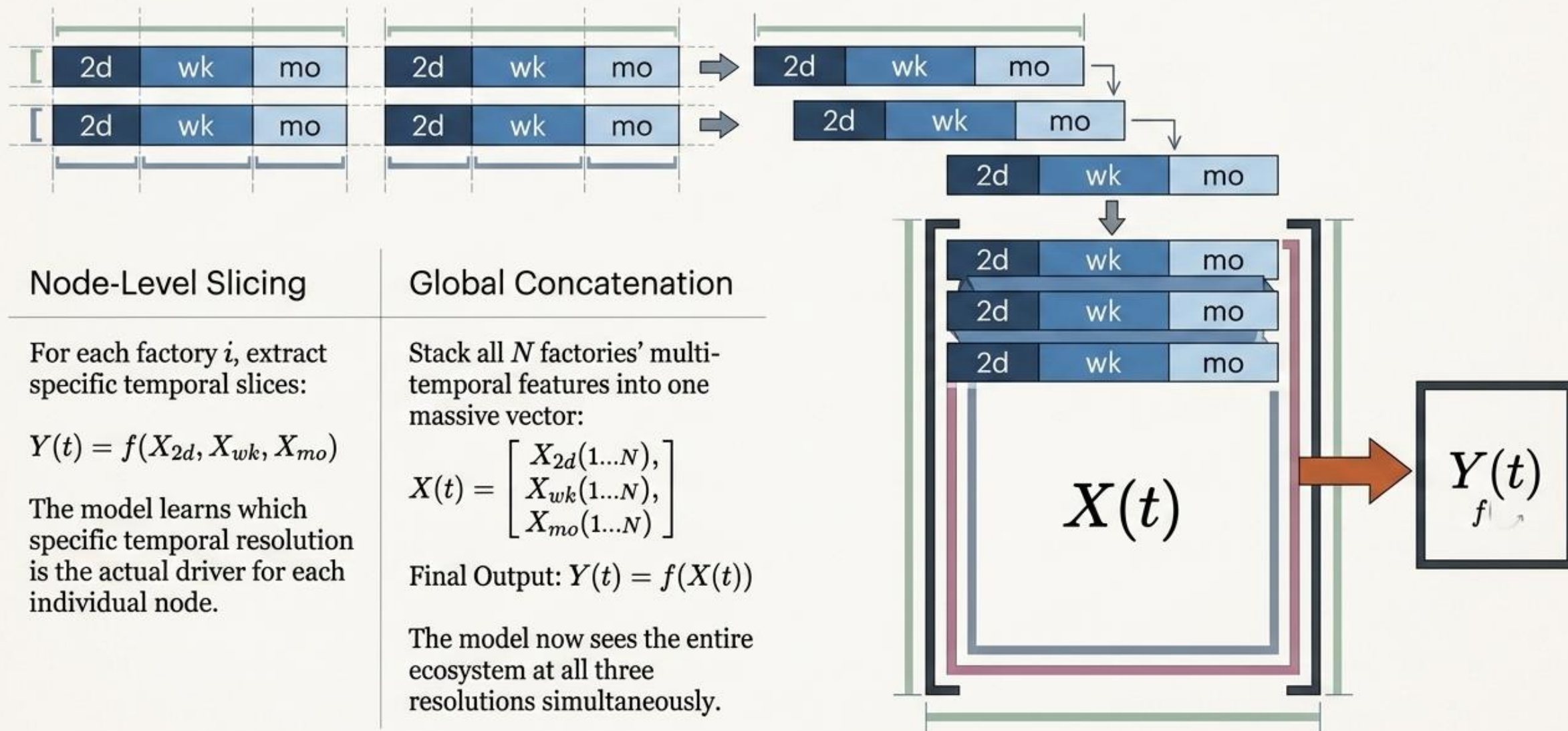
Seasonal baselines
Production trends
Long-run capacity

Solution: Multi-Temporal Resolution Regression



Instead of a single historical window, time is explicitly modeled at multiple resolutions simultaneously, allowing the algorithm to weight short, medium, and long-term histories differently for every single prediction.

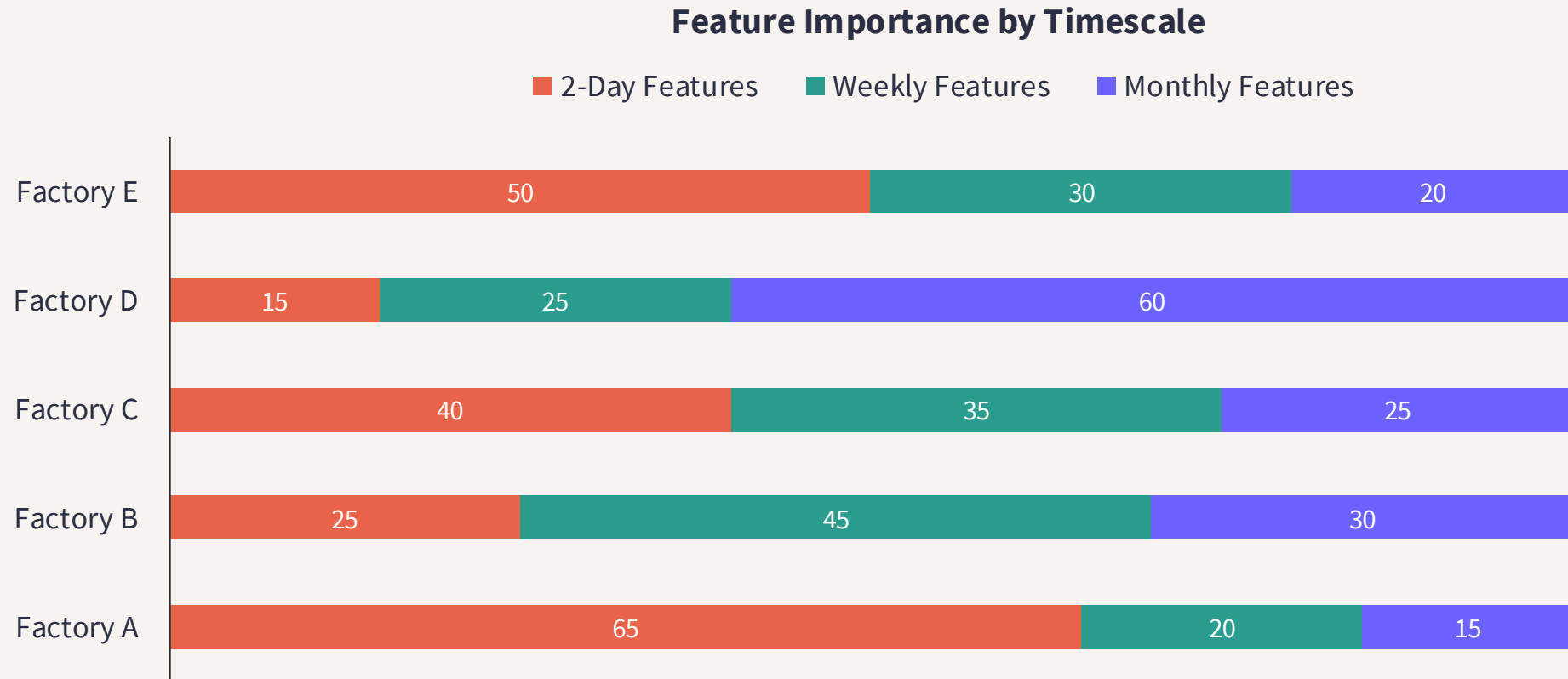
Constructing the Unified Feature Matrix



The Formulation: $Y(t) = f(X_{2d}, X_{wk}, X_{mo})$

- **Target $Y(t)$:** next-day peak power load (kW)
- **X_{2d} :** 48-hour rolling stats (mean, max, std of load)
- **X_{wk} :** 7-day rolling stats + day-of-week encoding
- **X_{mo} :** 30-day rolling stats + seasonal indicators
- All three concatenated per row → tree-based or linear model

What the Model Learns About Scale



Implication: A single-scale model misses factory-specific dynamics — multi-temporal lets the data decide which scales matter

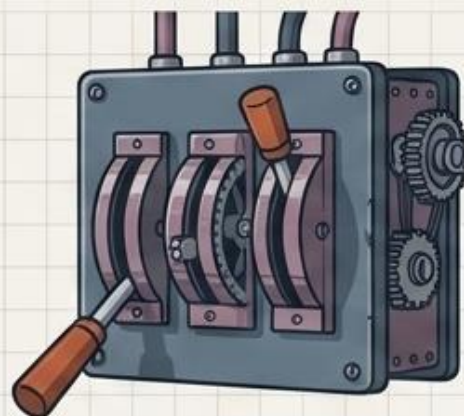
Section 2 Recap: Demand Forecasting

- Classical methods (ARIMA, Exp Smoothing) → univariate, quick baselines
- Prophet → strong seasonality + holidays with minimal tuning
- LSTM/DeepAR → complex nonlinear patterns if enough data
- Multi-temporal regression → let data choose the right timescale
- **Key lesson:** match method complexity to data richness

The Limits of Forecasting: Moving from What to Why

Forecasting relies on Association

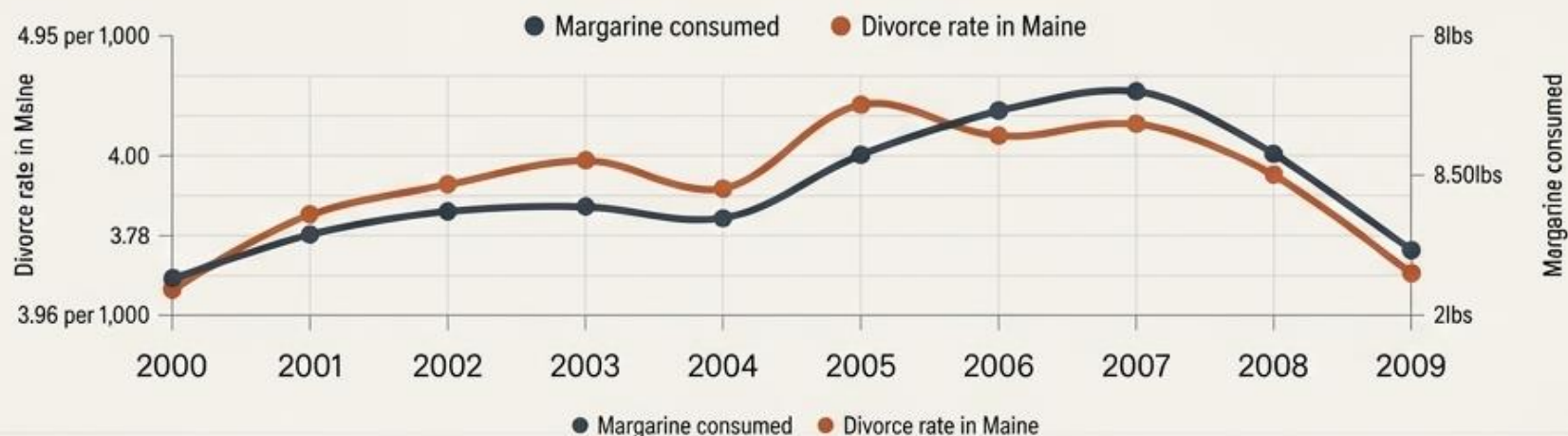
It tells you Y will go up, but it cannot tell you what lever to pull to change Y.



Causality allows for Intervention

To change the system, we must isolate predictive causation.

Divorce rate in Maine correlates with
Per capita consumption of margarine



The MDL Principle

Minimum Description Length dictates that true causes can compress the explanation of an effect, but not vice versa.

Forecasting models would blindly exploit this margarine correlation; a causal model rejects it.

Section 3

Variable-Lag Causality

VL-Granger Causality & Transfer Entropy

Amornbunchornvej, Chainarong, Elena Zheleva, and Tanya Y. Berger-Wolf.

The Definition: Predictive Precedence, Not Physical Intervention

True Causation

Nature: Deeply philosophical and physical.

Action: Requires physical intervention or counterfactuals.

Proof: "If I change X, Y will physically change."

Granger Causation

Introduced by Clive Granger, 1969

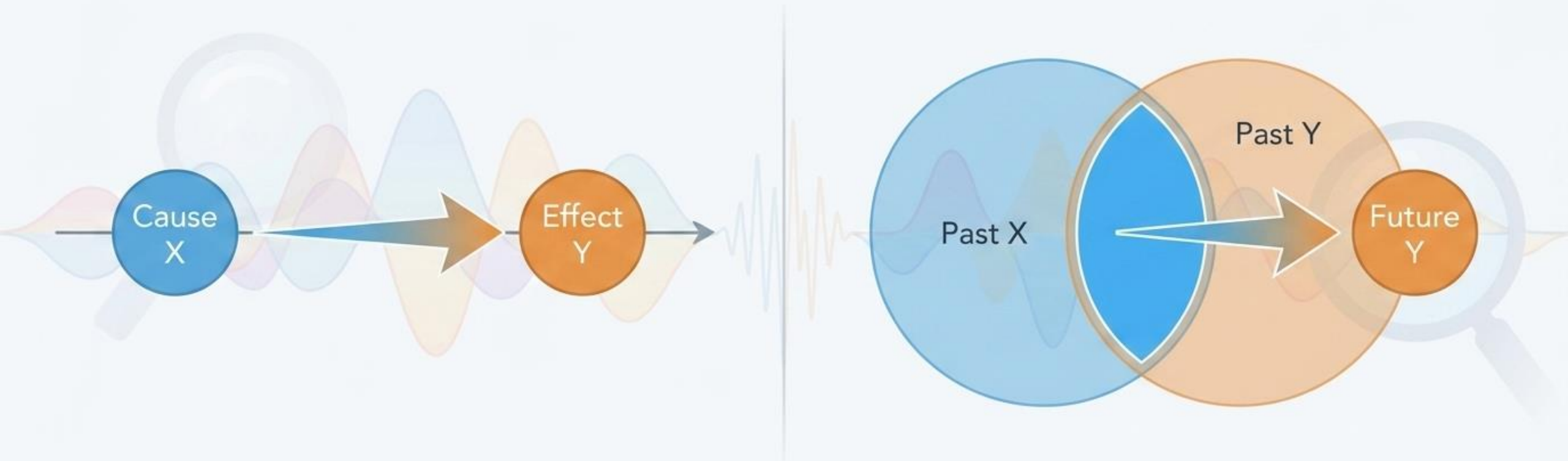
Nature: A statistical hypothesis test.

Action: Measures predictability and temporal precedence.

Proof: "Knowing the history of X helps me better forecast Y."

The term "causality" is an econometric misnomer. A more accurate description is **predictive precedence**. It avoids the post hoc ergo propter hoc fallacy by strictly measuring information gain.

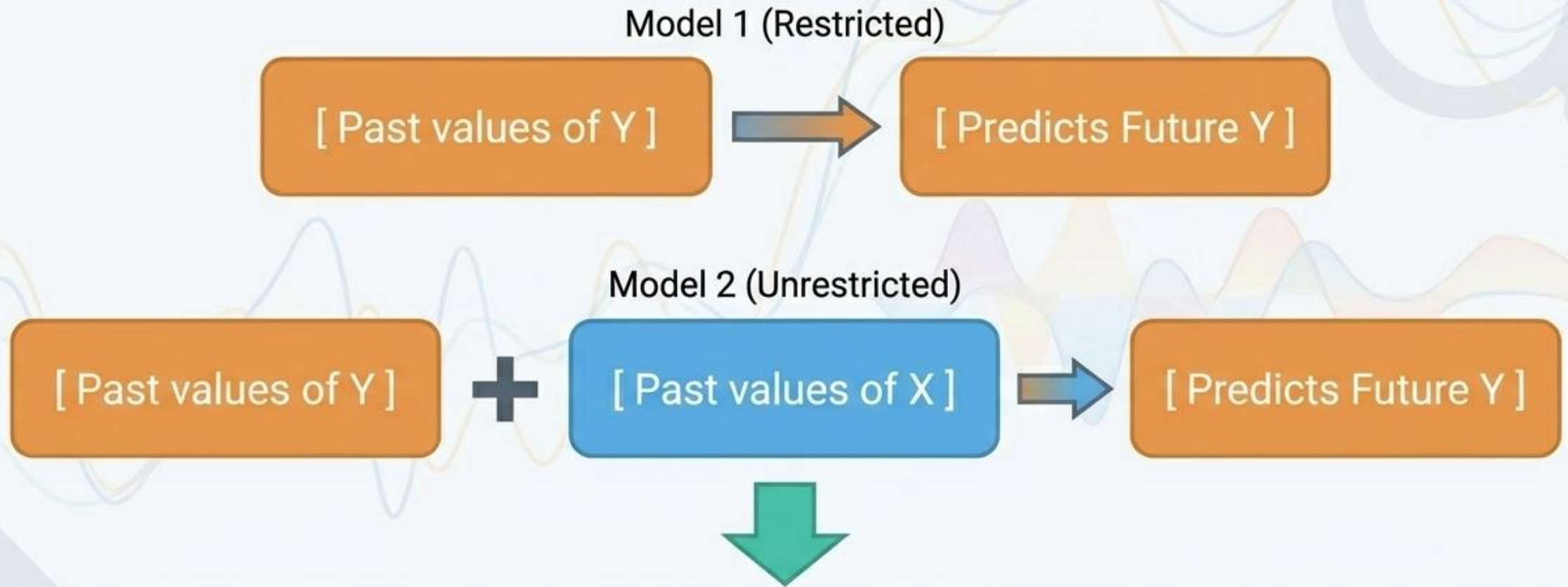
The Two Underlying Principles of Granger Causality



1. The cause happens prior to its effect.

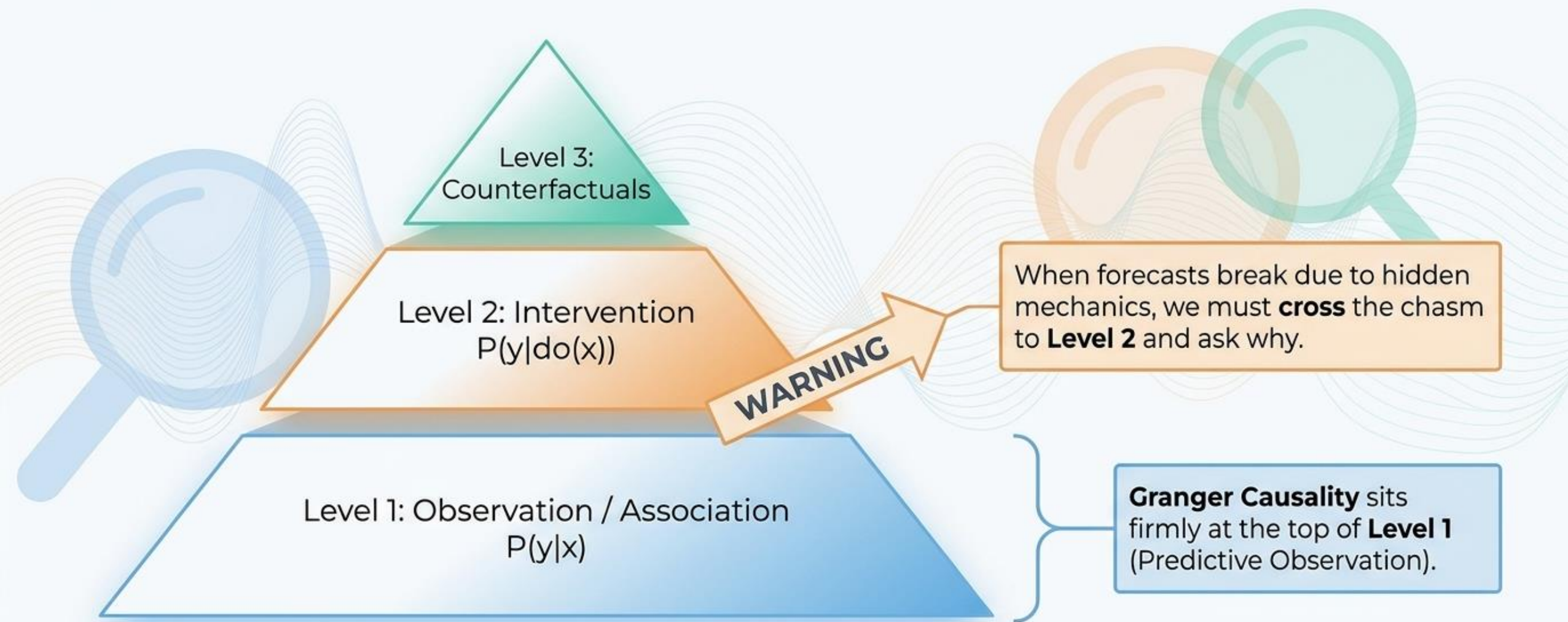
2. The cause contains unique information about the future values of its effect that is not present elsewhere.

The Mechanism: Vector Autoregression (VAR)



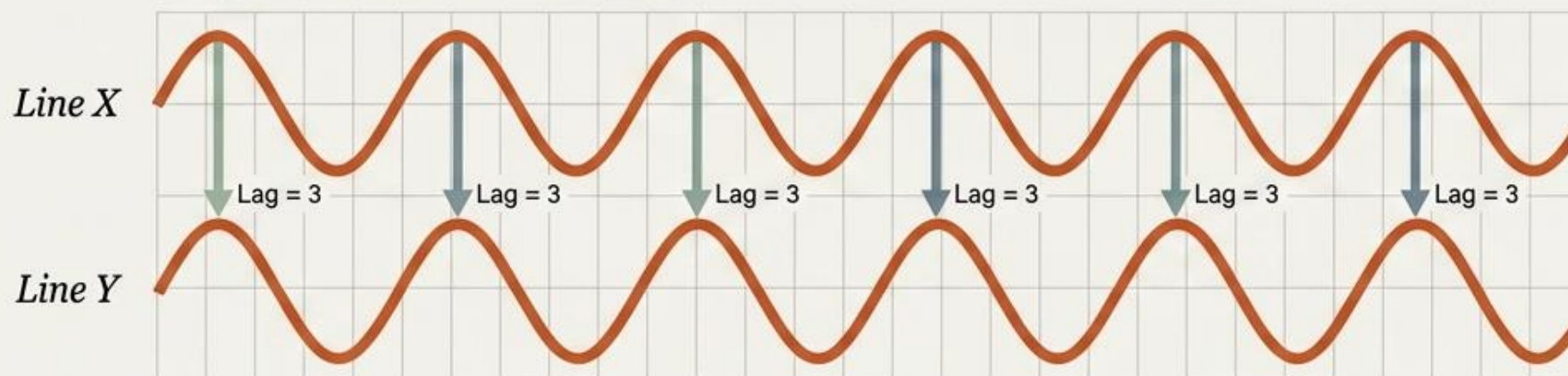
If Model 2 produces a statistically significant reduction in forecasting error compared to Model 1, then X Granger-causes Y.

The Causality Chasm



Standard Granger Causality and the Variable-Lag Problem

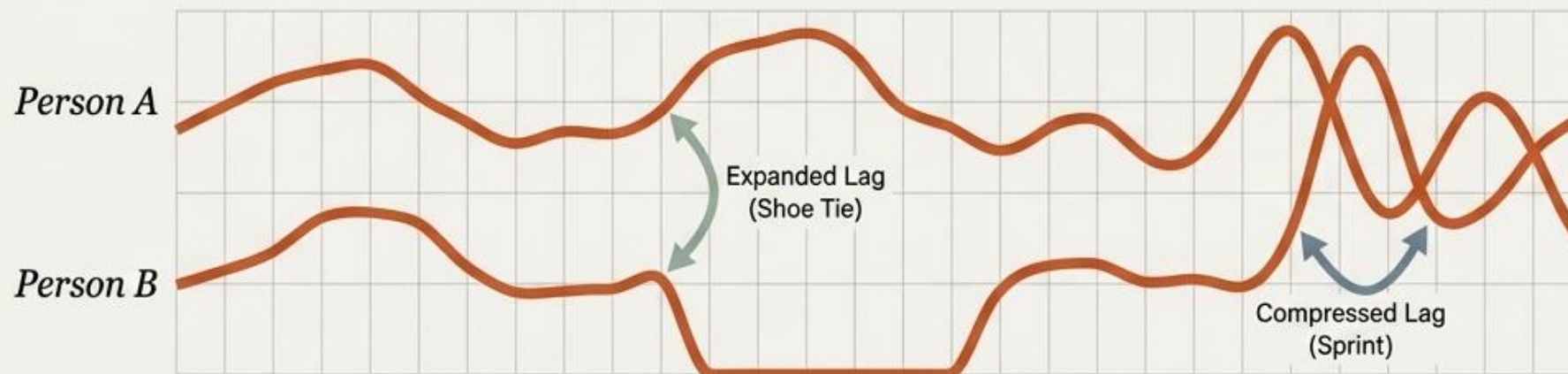
The Theory (Standard Granger)



C.W.J. Granger (1969):
X causes Y if past values of X better predict Y.

The fatal assumption: it assumes a fixed, constant metronome delay between cause and effect.

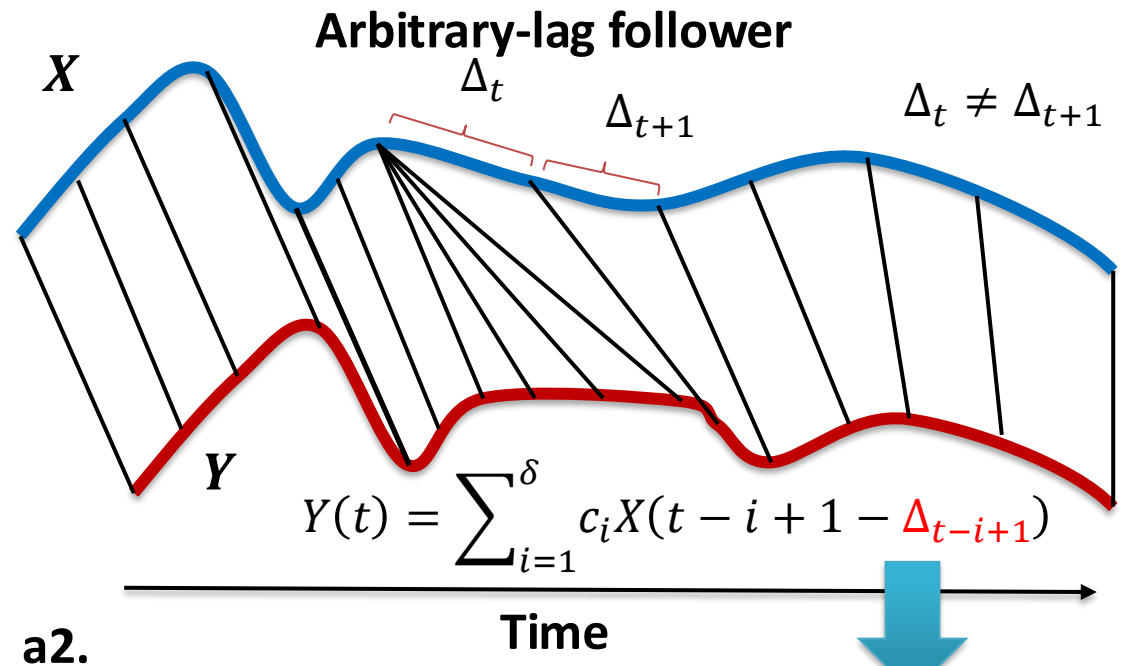
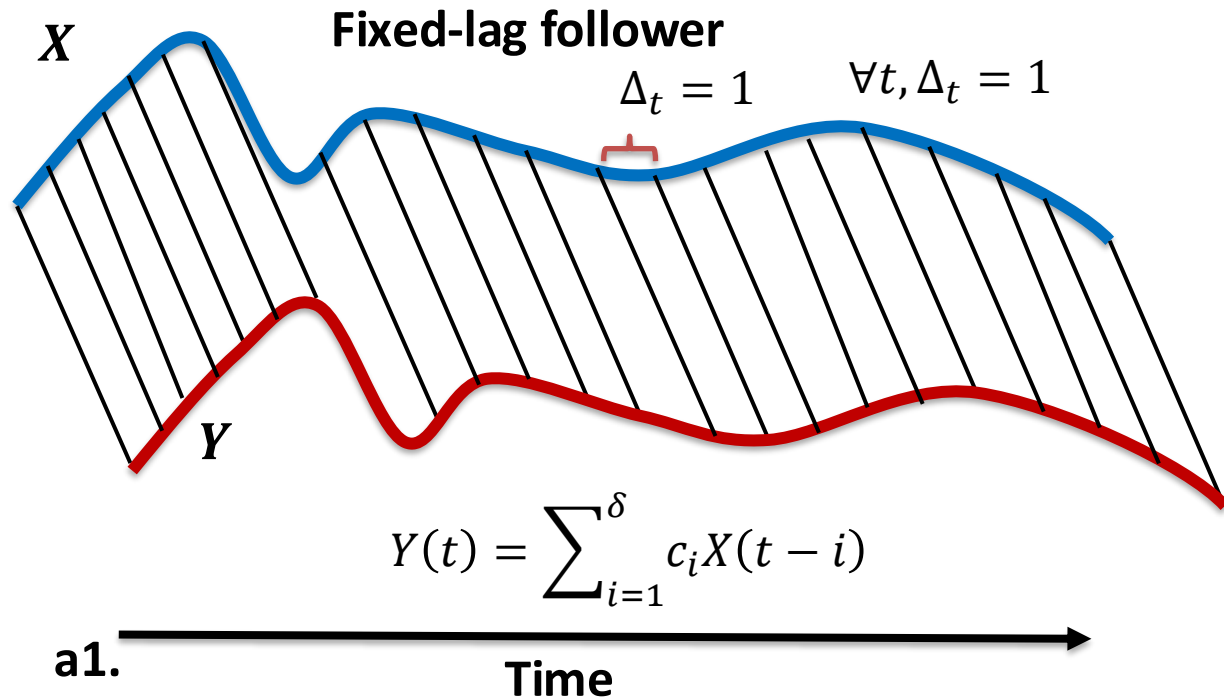
The Reality (Real-World Data)



Real-world time series are non-stationary. If a follower stops to tie their shoe, the delay expands and compresses.

Fixed-lag models completely break down.

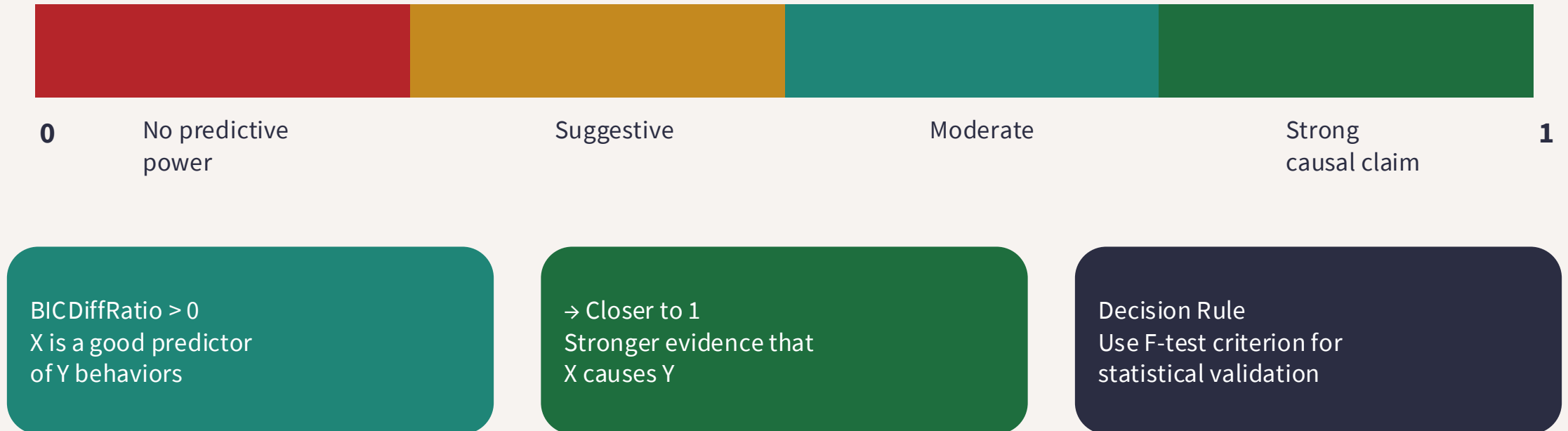
Fixed vs. Arbitrary lag follower



VL-Granger

BICDiffRatio: Measuring Causal Strength

Compares BIC of model with X as predictor vs. without X. Balances fit with complexity penalty.

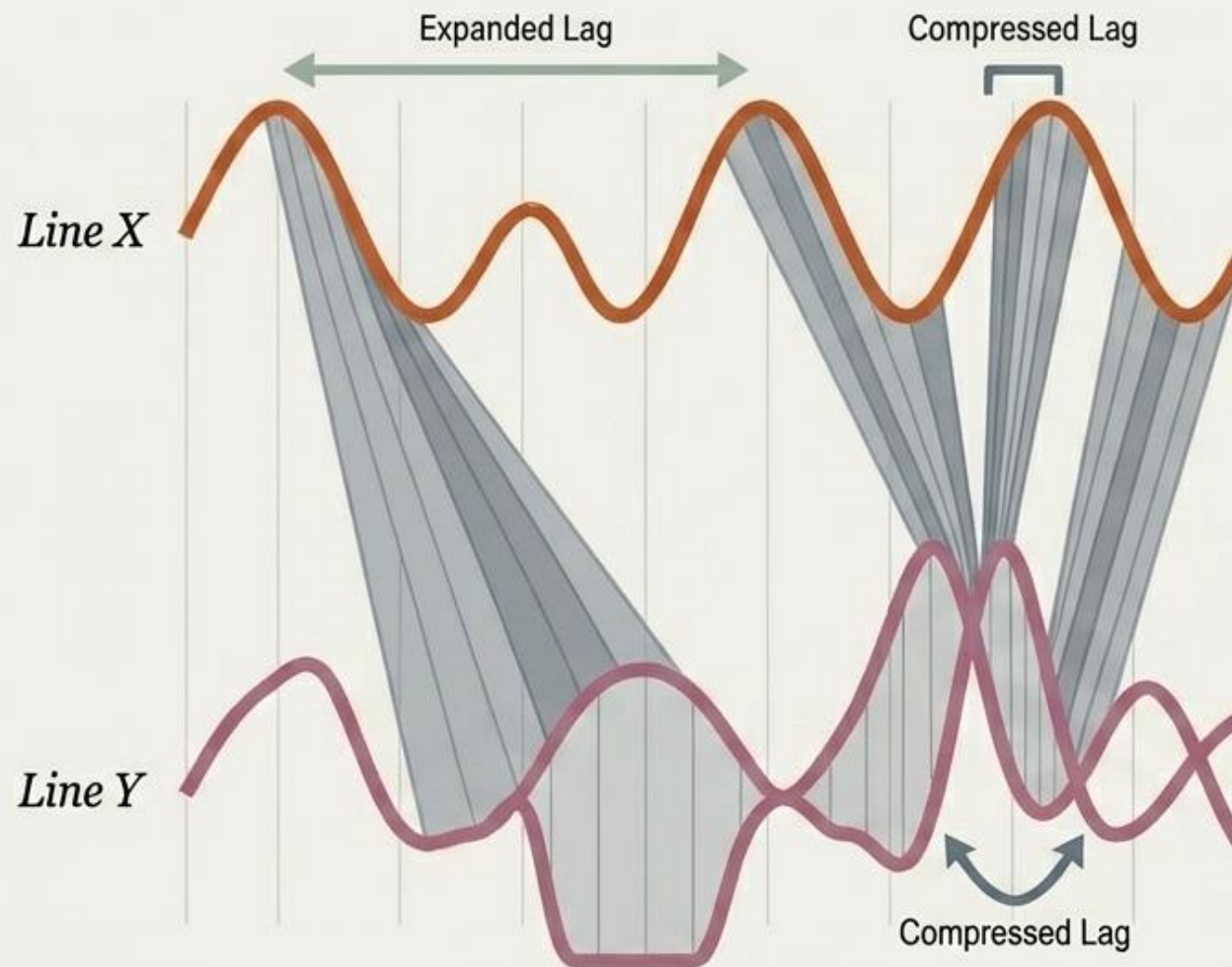


Advantage: balances model fit with complexity penalty — avoids overfitting

The Solution: VL-Granger Causality

Dynamic Alignment

Dynamic Time Warping (DTW) stretches and compresses the time axis to structurally align the candidate cause X to the effect Y , regardless of fluctuating delays.



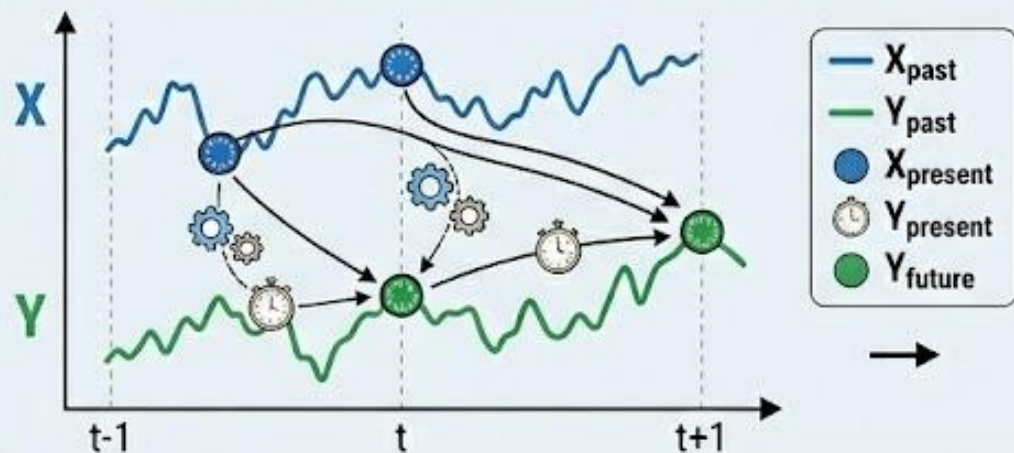
Implementation via VLTimeCausality

Once structurally aligned, a Granger-style residual test is applied to the warped data using the F -test decision criterion.

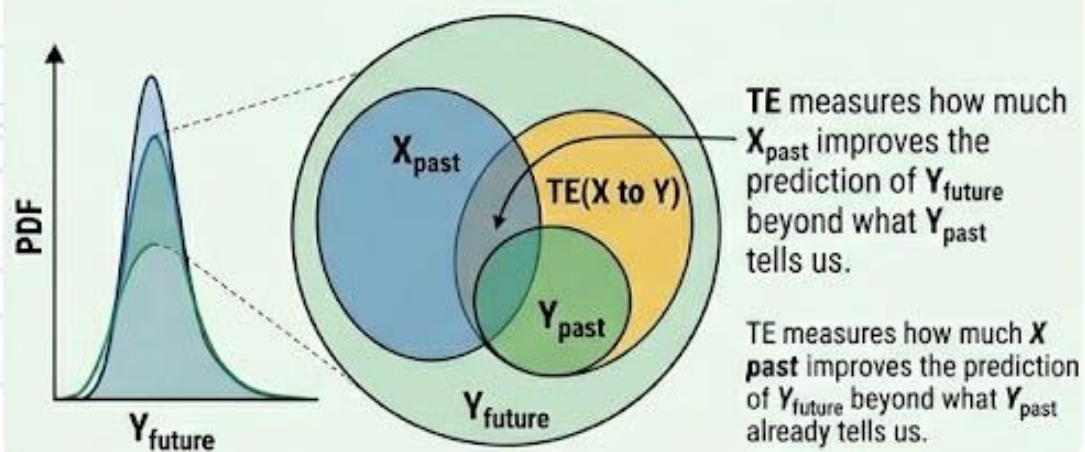
We evaluate the *BICDiffRatio*: if >0 , X is a robust predictor of Y 's behavior.

VISUALIZING TRANSFER ENTROPY (TE) IN TIME SERIES ANALYSIS

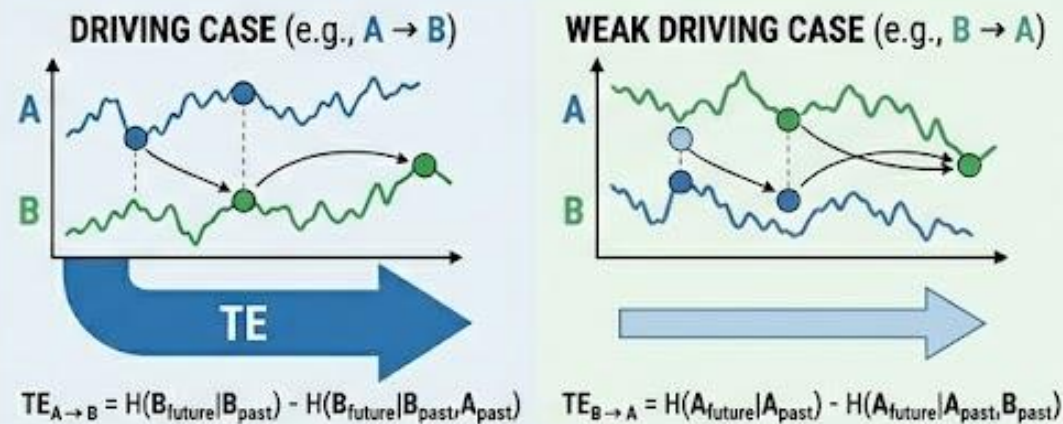
STEP 1: THE COMPONENTS AND RELATIONSHIPS



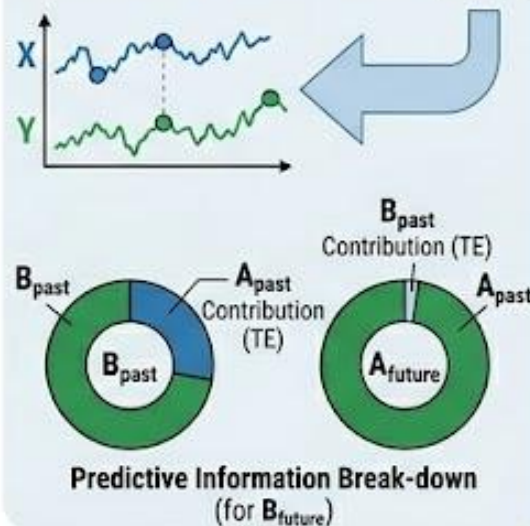
STEP 2: REDUCING UNCERTAINTY IN Y



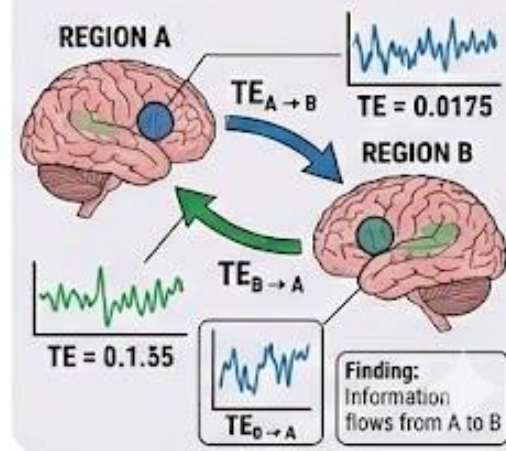
STEP 3: DIRECTED INFORMATION FLOW IS ASYMMETRIC



RESPONDING RELATIONSHIP ($Y \rightarrow X$)



STEP 4: REAL-WORLD APPLICATION EXAMPLE



VL-Transfer Entropy

Information-Theoretic

- Extends Transfer Entropy with variable lags via DTW
- Captures nonlinear dependencies
- Model-free: no assumption on data distribution

When to Use Which?

- **VL-Granger:** linear relationships, fast computation
- **VL-TE:** nonlinear or complex interactions
- Both available in the R package

VLTimeCausality: R Package

```
library(VLTimeCausality)

# Generate simulation data
TS <- SimpleSimulationVLtimeseries()

# Test VL-Granger causality
out <- VLGrangerFunc(Y=TS$Y, X=TS$X)

# Results
out$XgCsY          # TRUE/FALSE
out$BICDiffRatio   # 0 to 1
```

Key Outputs

XgCsY
Boolean causal decision

BICDiffRatio
0 = no causality
1 = strong causality

F-test p-value
Significance check

Real-World Applications

Collective Behavior



Animal Movement
Coordinated flocking
and migration patterns



Social Influence
Leader-follower dynamics
in social networks



Brain Connectivity
fMRI neural region
causal pathways

Industrial / Financial



Supply Chains
Disruption propagation
across suppliers



Stock Markets
Lead-lag relationships
between assets



Industrial IoT
Sensor-to-sensor
causal diagnostics

Section 3 Recap: Variable-Lag Causality

1

DTW Alignment

The bridge to variable-lag causality

2

BICDiffRatio

Continuous strength measure from 0 to 1

3

VL-Transfer Entropy

Handles nonlinear dependencies

4

Multi-Band Extension

Adds frequency-domain analysis

R

Open Source

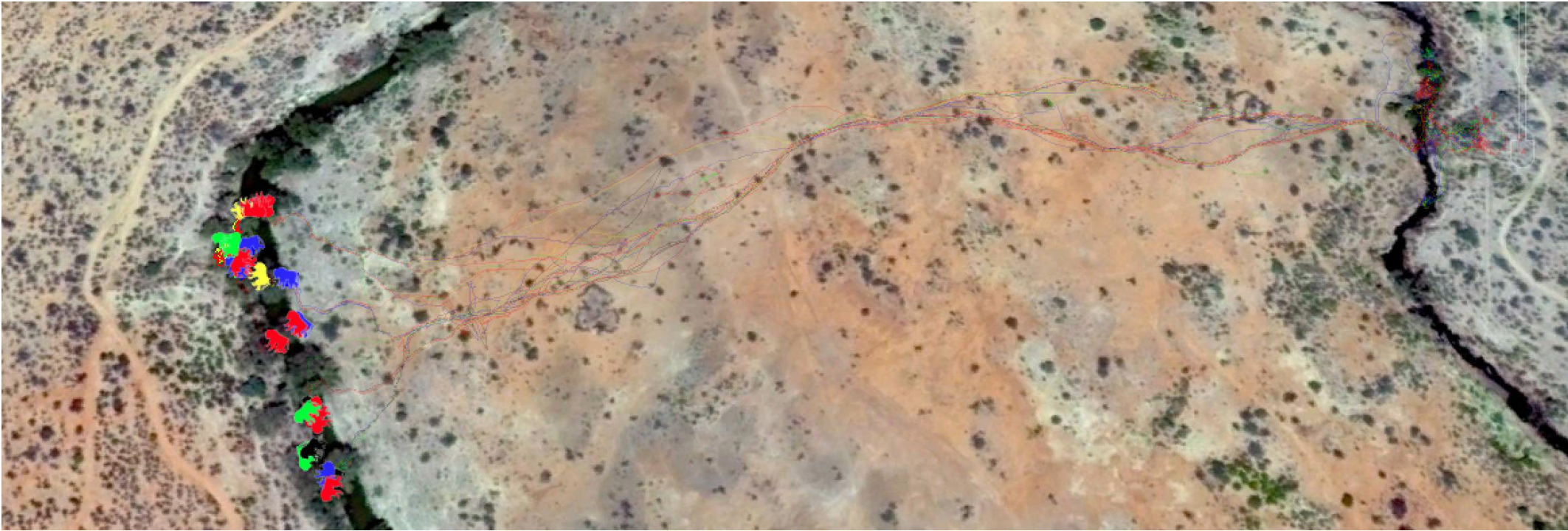
VLTimeCausality on R-CRAN

Section 4

Coordination & Leadership

mFLICA: Multi-Faction Leadership Inference from Coordinated Activities
Amornbunchornvej, Chainarong and Tanya Y. Berger-Wolf.

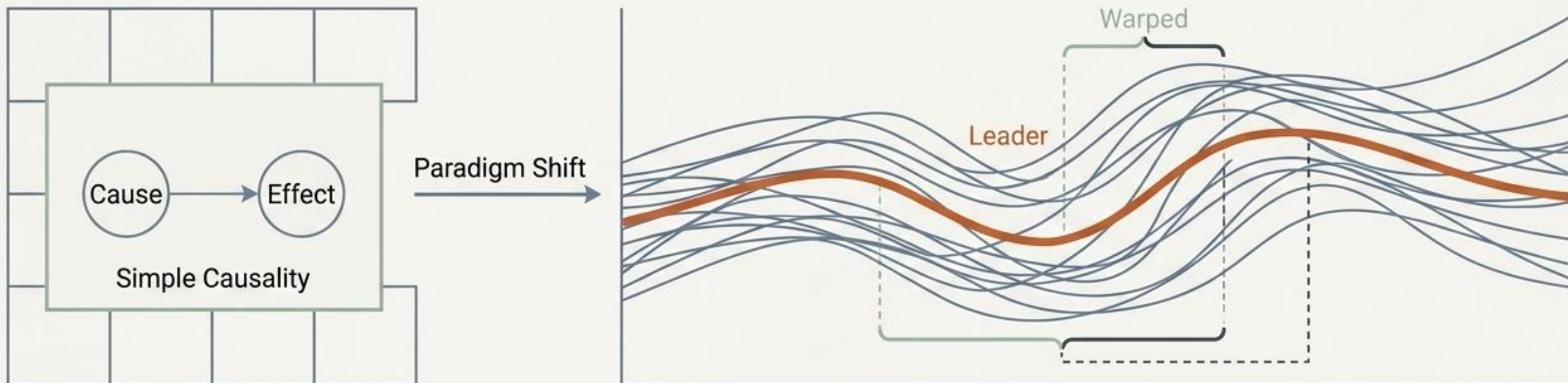
Time series: Baboon movement (via GPS-collar tracking)



INPUT: Given data points $x_1, x_2, \dots$ of some system in the past

GOAL: Who is a leader whom everyone follows in collective movement?

Beyond Forecasting: Group Dynamics and mFLICA



The Paradigm Shift

Moving from predicting a single factory to tracking multiple moving agents. Time series encode unseen social and hierarchical structures.

The Leader-Follower Problem

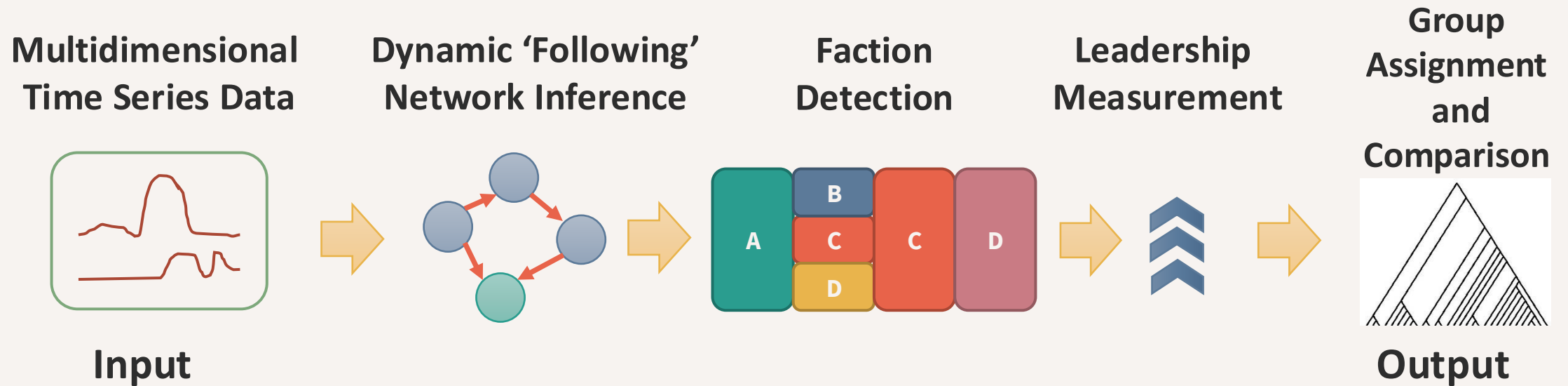
How do we detect factions and coordination in entirely unlabelled time series purely from temporal movement data?

Inference via mFLICA

Multi-faction leader-follower dynamics infer who is leading without requiring explicit interaction data.

Applied to animal flocking, social media coordination, or financial market followers.

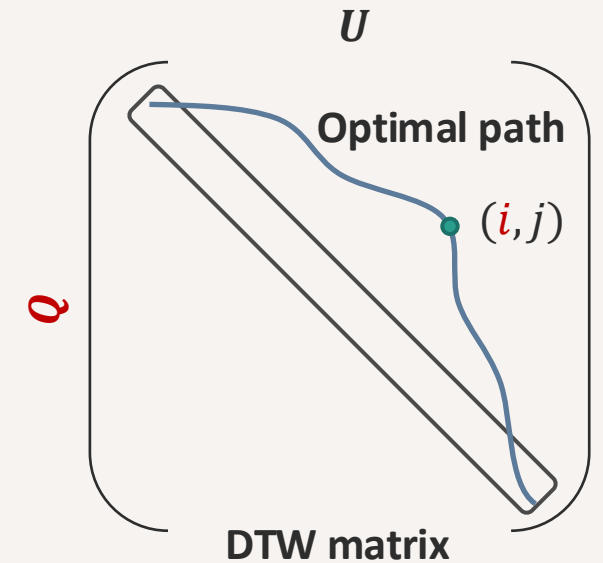
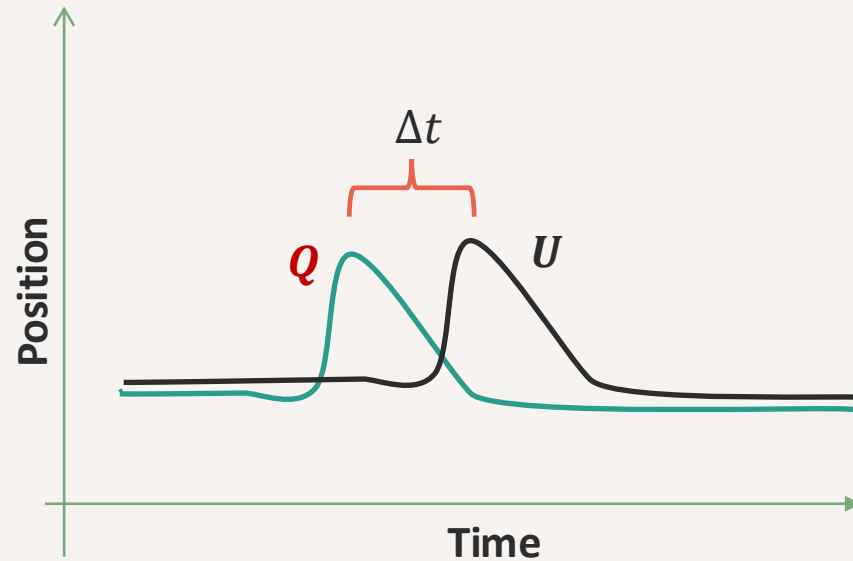
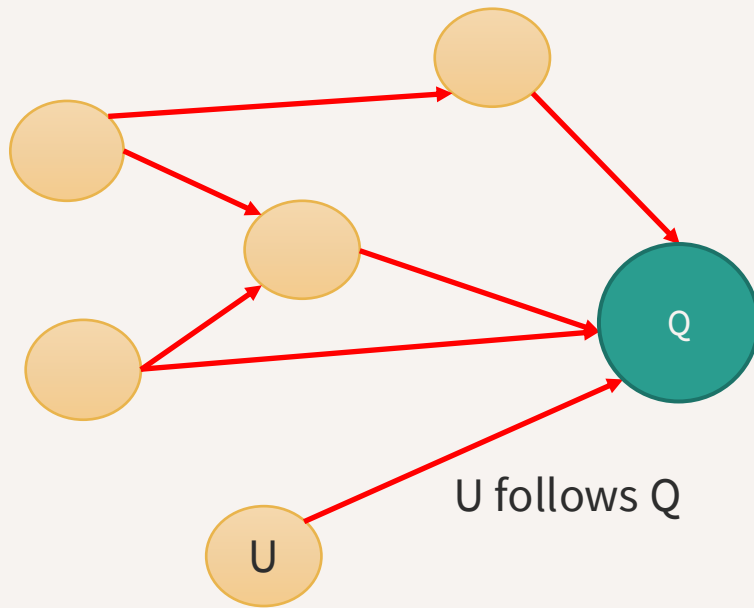
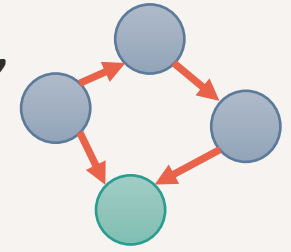
High-level mFLICA* framework procedure



*The Framework of Multiple-Faction Leadership Inference in Coordinated Activity

Following Network

Dynamic 'Following'
Network Inference

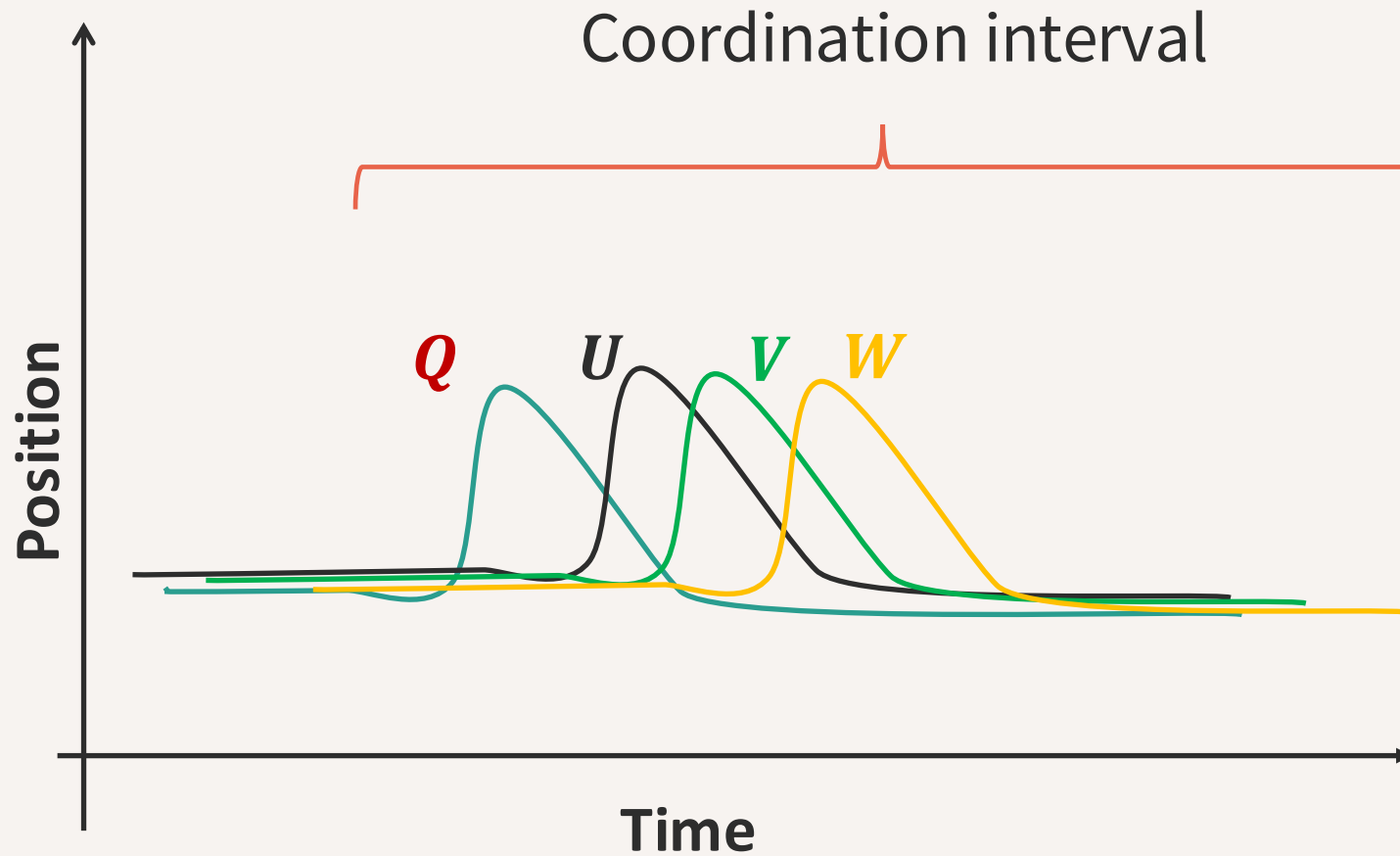


Following Network
Nodes: individuals
Edges: following relations

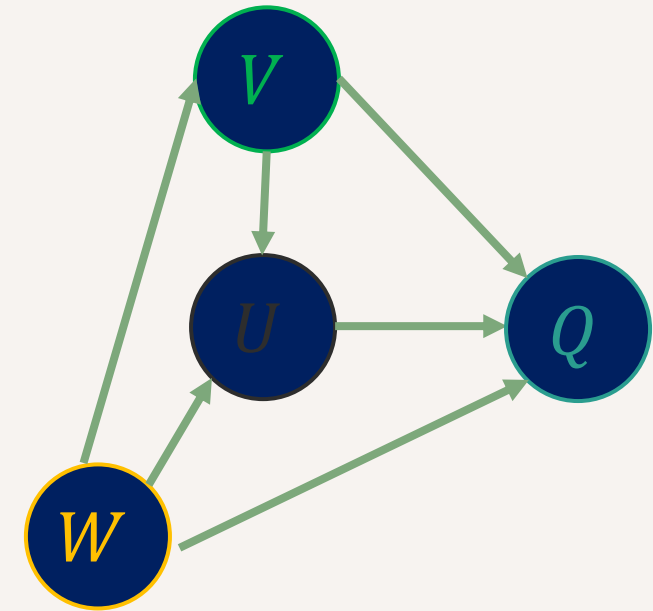
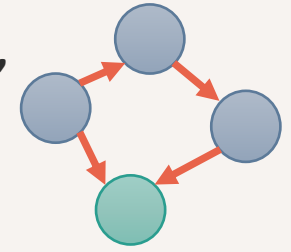
Time Series of U follows Q

The optimal warping
path of DTW

Coordination



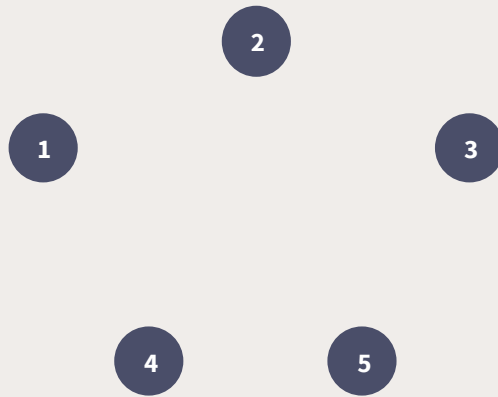
Dynamic 'Following'
Network Inference



Following Network

Leader & Faction Detection

Following Relations



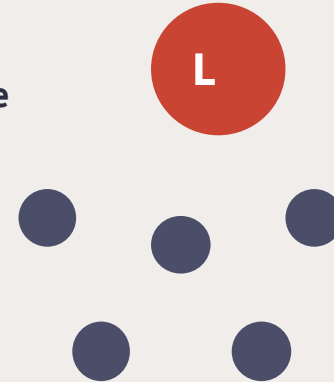
DTW tests each pair (i, j) for variable-lag following

Builds a directed following network at each time step

Edge $(i \rightarrow j)$ means "i follows j"

Leader & Faction Detection

High in-degree
Zero out-degree

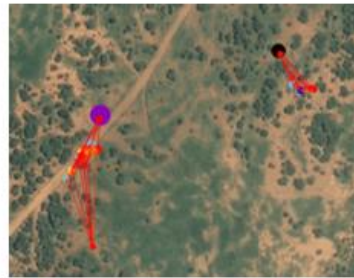
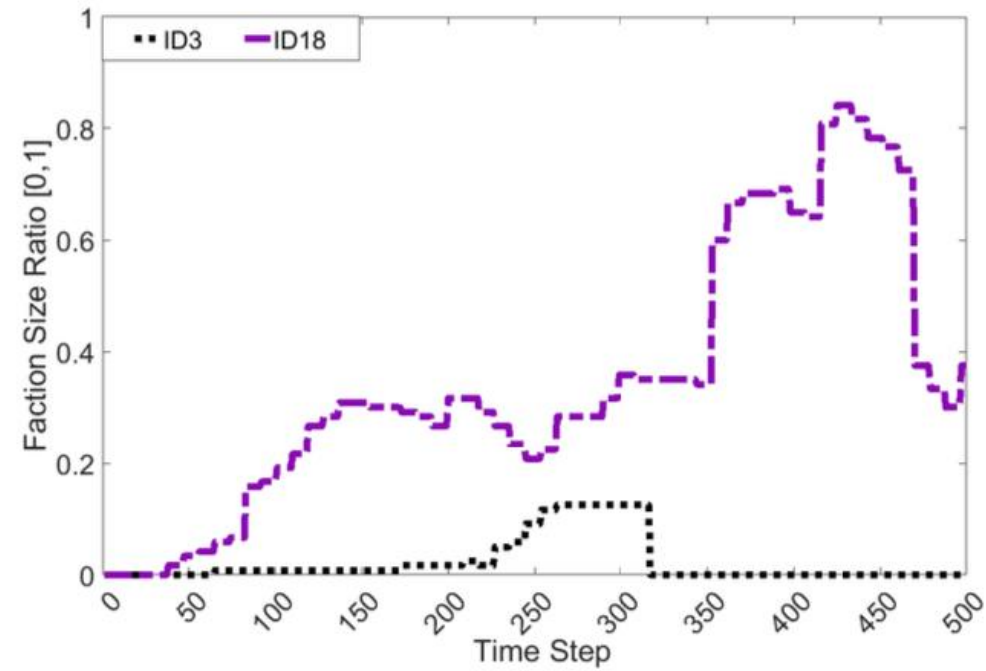


Leader: initiates movement everyone follows but follows no one

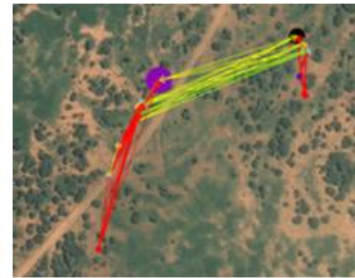
Faction: group of individuals following the same leader

Multiple factions can coexist simultaneously

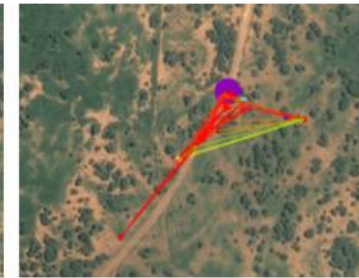
Case study: merging event of baboon troops.



(b) $t = 300$



(c) $t = 350$



(d) $t = 400$

mFLICA R Package

```
# Install
install.packages("mFLICA")
library(mFLICA)

# Load 30-individual dataset
# TS[i,t,d]: individual i, time t, dim d

# Run mFLICA analysis
result <- mFLICA(mFLICA::TS)

# Get leaders & factions
result$leaders
result$factions
```

Outputs

Following relations
Leaders of factions
Faction members
Time-step-level results

Publications

Amornbunchornvej & Berger-Wolf, SIAM
SDM 2018
SoftwareX 2021
CRAN: mFLICA

Section Recap: Coordination



When, Who, How

mFLICA answers coordination in multivariate time series



DTW Pairwise Following

Builds directed networks from variable-lag alignment



Leader Detection

High in-degree + zero out-degree identifies initiators



R-CRAN Package

Built-in 30-individual benchmark dataset included



Next Up

Decomposition and closing thoughts

Section 5

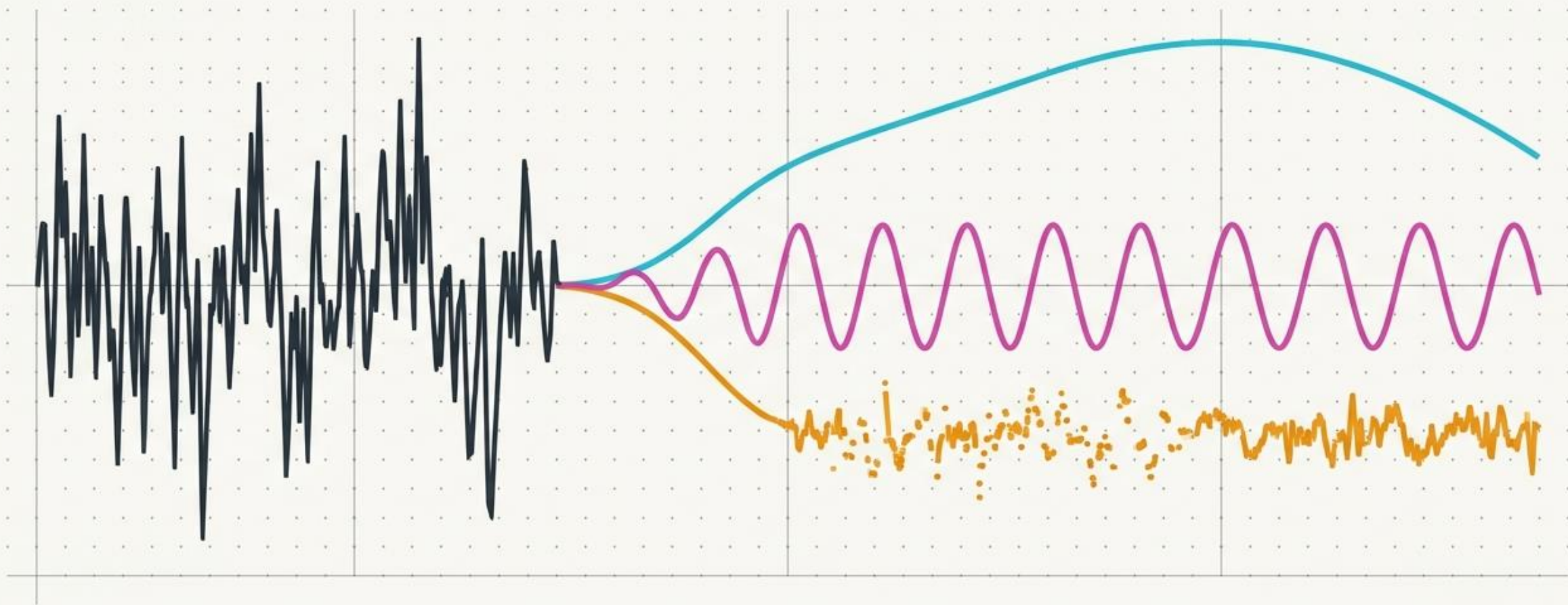
Decomposition & Closing

LGTD, Connecting the Toolkit, and Future Directions

Sophaken, Chotanansub, Thanadej Rattanakornphan, Piyanon Charoenpoonpanich, Thanapol Phungtua-eng, and Chainarong Amornbunchornvej

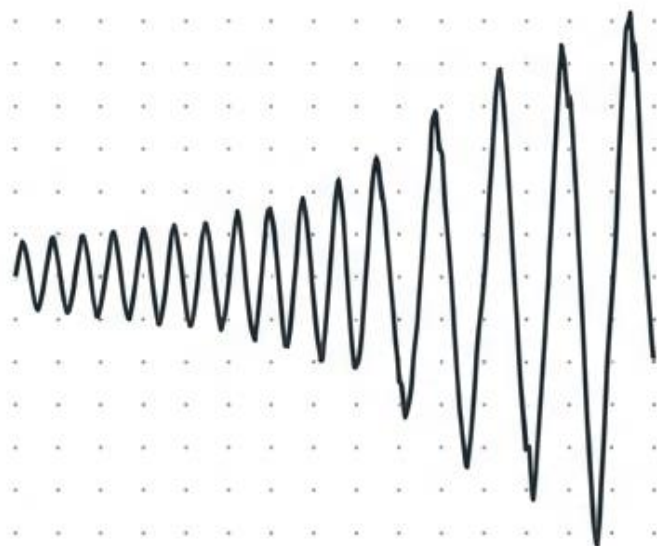
LGTD: Local-Global Trend Decomposition

A principled, season-length-free approach to time series analysis.



Open-Source Implementation (Python / Java) | MIT License

Non-Stationary Seasonality



Cycles that stretch, compress, or change intensity.

Intermittent Patterns



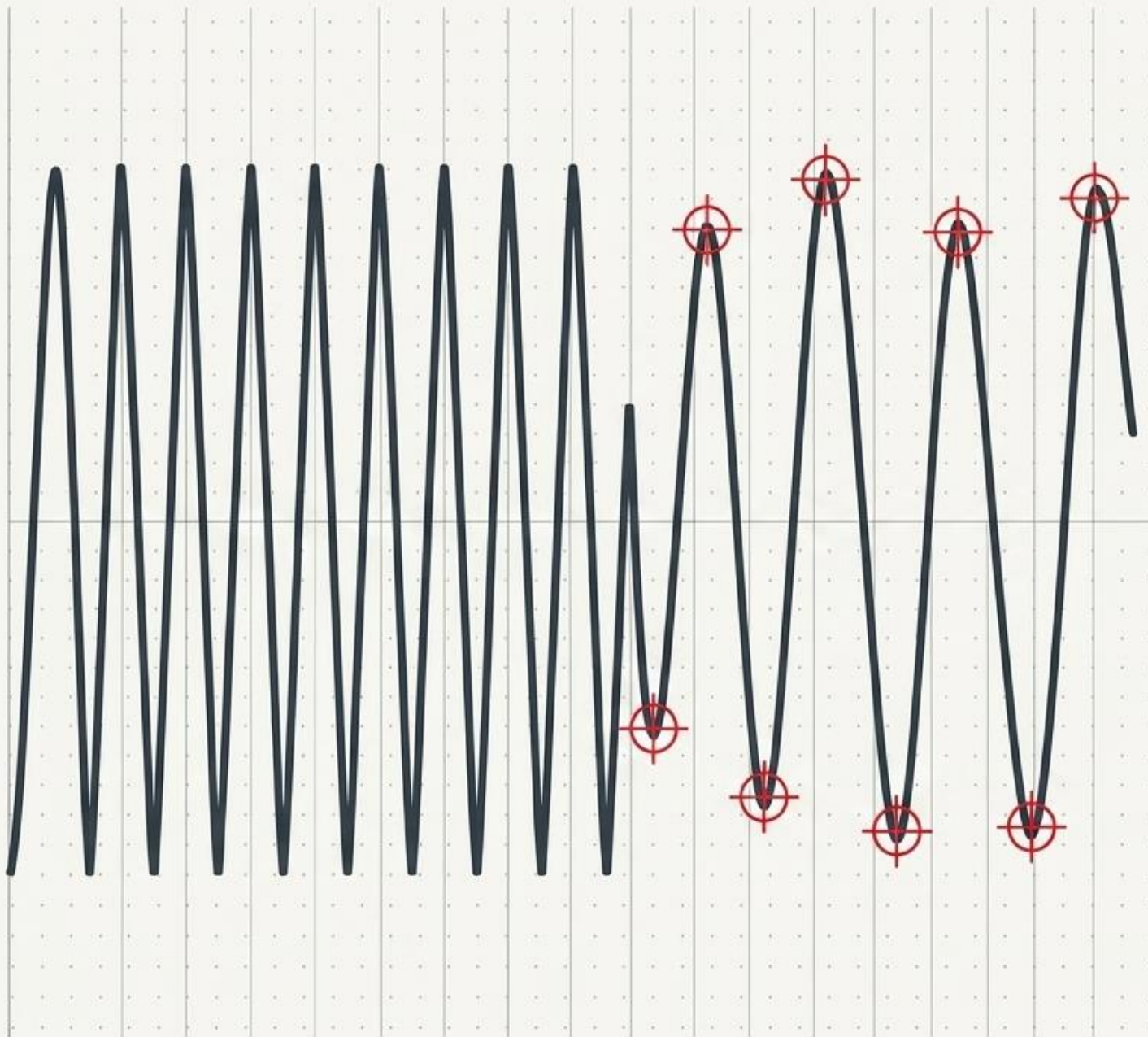
Seasonal behaviors that pause and resume unpredictably.

Multiple Overlapping Cycles



Interacting periodicities that drift in and out of phase.

Real-world time series rarely follow textbook rules. When seasonality drifts, overlaps, or disappears, classical assumptions fail.



The Bottleneck of Classical Methods

Classical decomposition techniques—like STL, X-11, and MSTL—are inherently period-dependent.

The Flaw: They require fixed or user-specified seasonal lengths.

The Consequence: When periods overlap or drift in real-world data, these models become brittle. Incorrect or misspecified periods substantially degrade decomposition quality.

The LGTD Paradigm Shift

LGTD eliminates the need for explicit period selection. It automatically identifies and aggregates repeating structures of arbitrary and potentially time-varying scales.

$$y_t = T_t + S_t + R_t$$



Global trajectory
(Adaptive)



Aggregated structure
(Season-Length-Free)



Isolated noise
(Robust)

The Evolution of Decomposition

Feature	Classical (STL / X-11)	LGTD
Seasonal Assumption	Fixed	Season-Length-Free
Period Input	Required / User-specified	Fully Automatic
Drifting Cycles	Degrades / Fails	Seamlessly Supported
Trend Modeling	Rigid	Adaptive (Linear / LOWESS)

LGTD shifts time series decomposition from a manual, trial-and-error tuning process into a fully automated extraction engine.

Core Capabilities Framework

ROBUSTNESS

SEASON-LENGTH-FREE

Makes no prior assumptions of known or fixed seasonal periods.

ADAPTIVE TREND MODELING

Automatically selects the optimal approach between linear and LOWESS trends.

MULTI-PERIODICITY

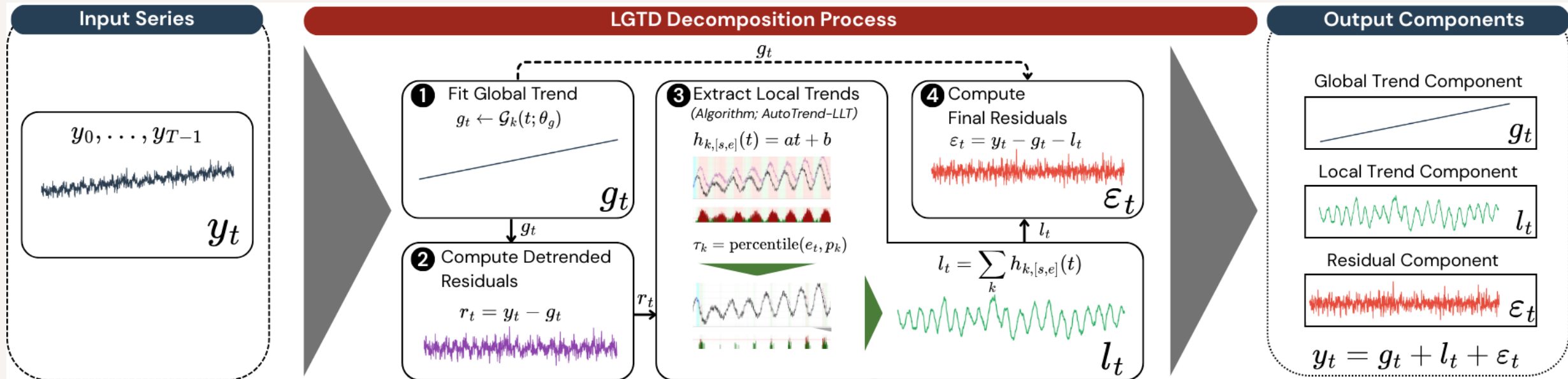
Seamlessly tracks multiple overlapping and time-varying periodicities.

NOISE ROBUSTNESS

Maintains stable performance under irregular sampling and highly noisy observations.

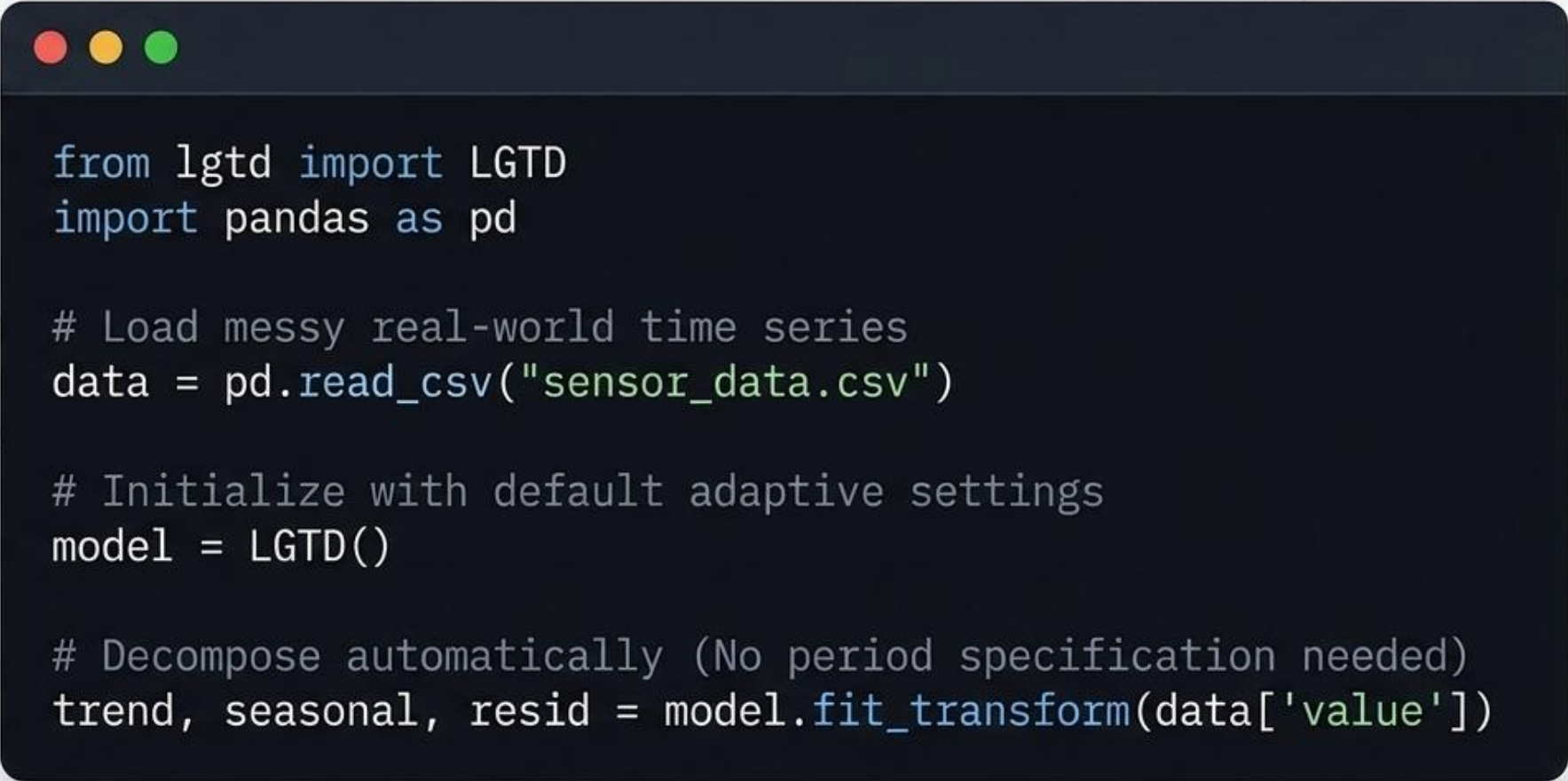
FLEXIBILITY

How it works



Built for Practitioners

Minimal Interface. Powerful Backend. LGTD features a familiar **scikit-learn-style** API for immediate integration into existing Python data science workflows.



```
from lgtd import LGTD
import pandas as pd

# Load messy real-world time series
data = pd.read_csv("sensor_data.csv")

# Initialize with default adaptive settings
model = LGTD()

# Decompose automatically (No period specification needed)
trend, seasonal, resid = model.fit_transform(data['value'])
```

Empirical Rigor & Reproducibility

7 Baselines

Rigorously evaluated against seven industry-standard baseline decomposition methods.

Synthetic Data

Tested against controlled environments to mathematically prove multi-periodicity handling.

Real-World Data

Validated on complex, noisy real-world datasets to ensure production readiness.

The repository contains a complete, automated experimental pipeline for reproducing all results reported in the accompanying paper.

Ready to Deploy

chotanansub/LGTD

Language: Python (90.9%) | Java (9.1%)

License: MIT

Paper: arxiv.org/abs/2601.04820

Docs: Includes full API Reference, Parameter Guides, and mathematical formulation.

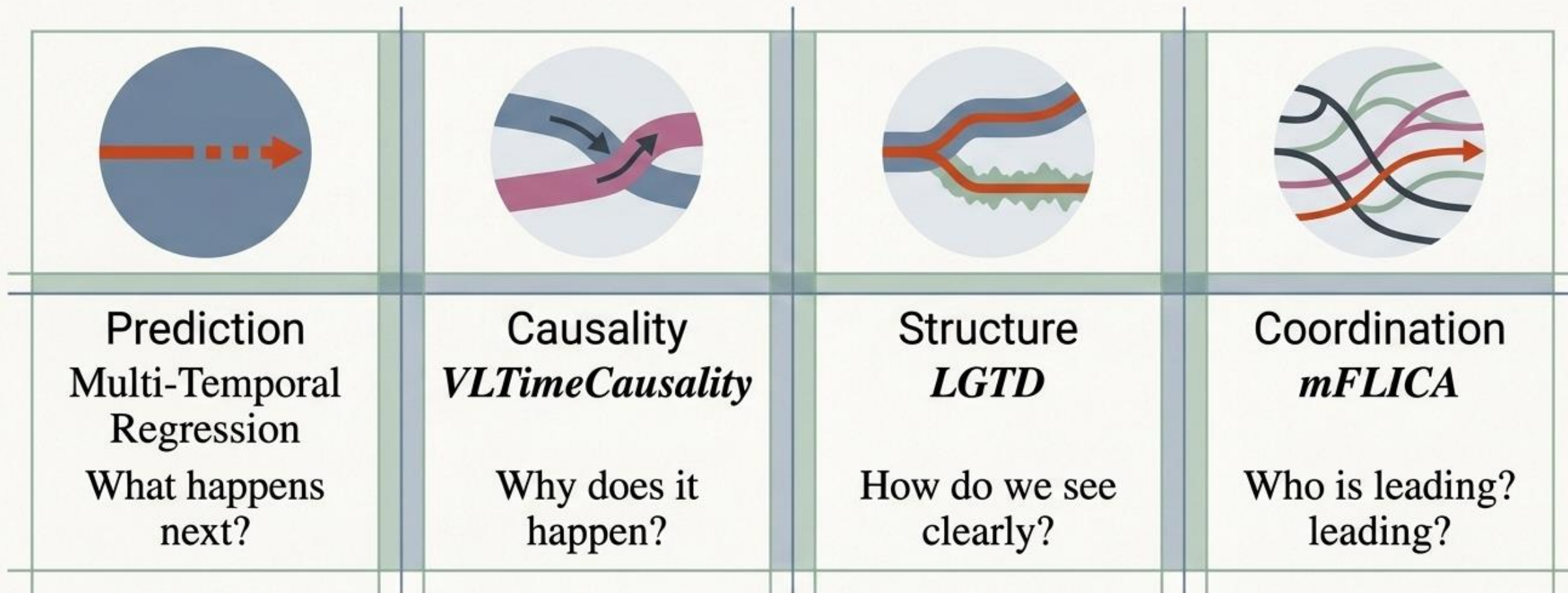
Experiments: Full reproducibility instructions and dataset access included in the repo.

Stop manually tuning fixed seasonal periods. Deploy LGTD today.

Future Directions

- **Multi-Band VL-Granger Causality** extends to frequency-domain causal analysis
- Foundation models for time series (TimeGPT, Chronos) as forecasting baselines
- Combining LGTD's decomposition with causal features as pre-processing for deep models
- Online/streaming versions of mFLICA for real-time coordination detection
- Cross-domain transfer: can causal structures learned in one factory generalize to another?

The Time Series Ecosystem



Stop asking: "How do I predict this?"
Start asking: "What hidden relationship am I actually trying to uncover?"